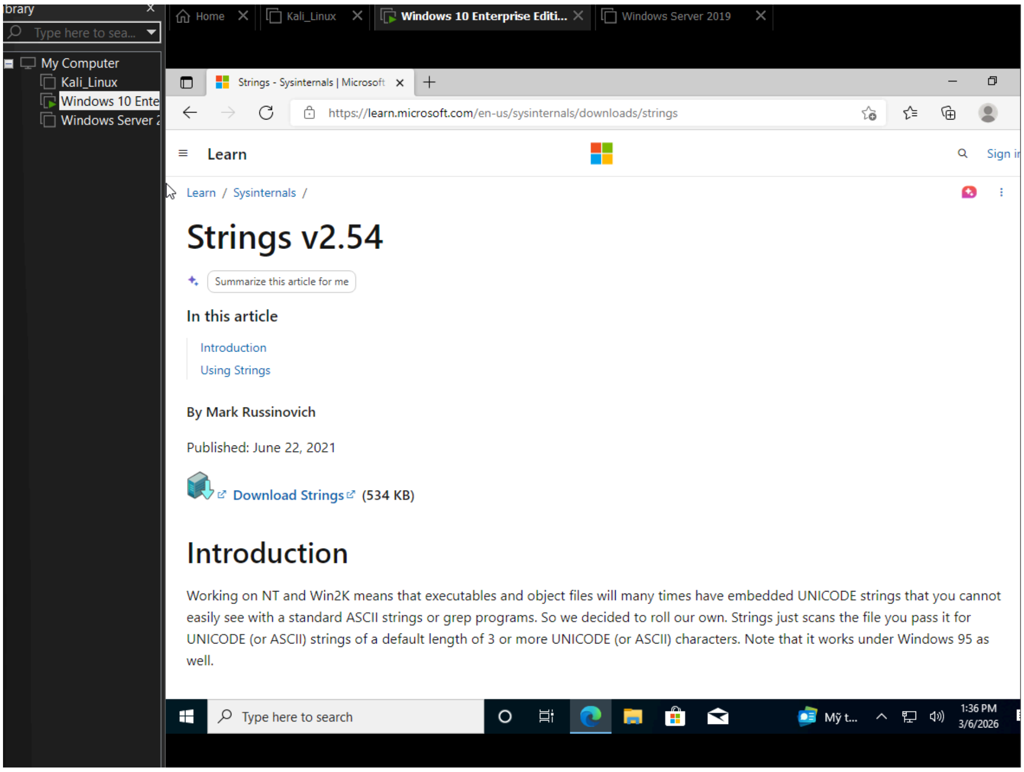
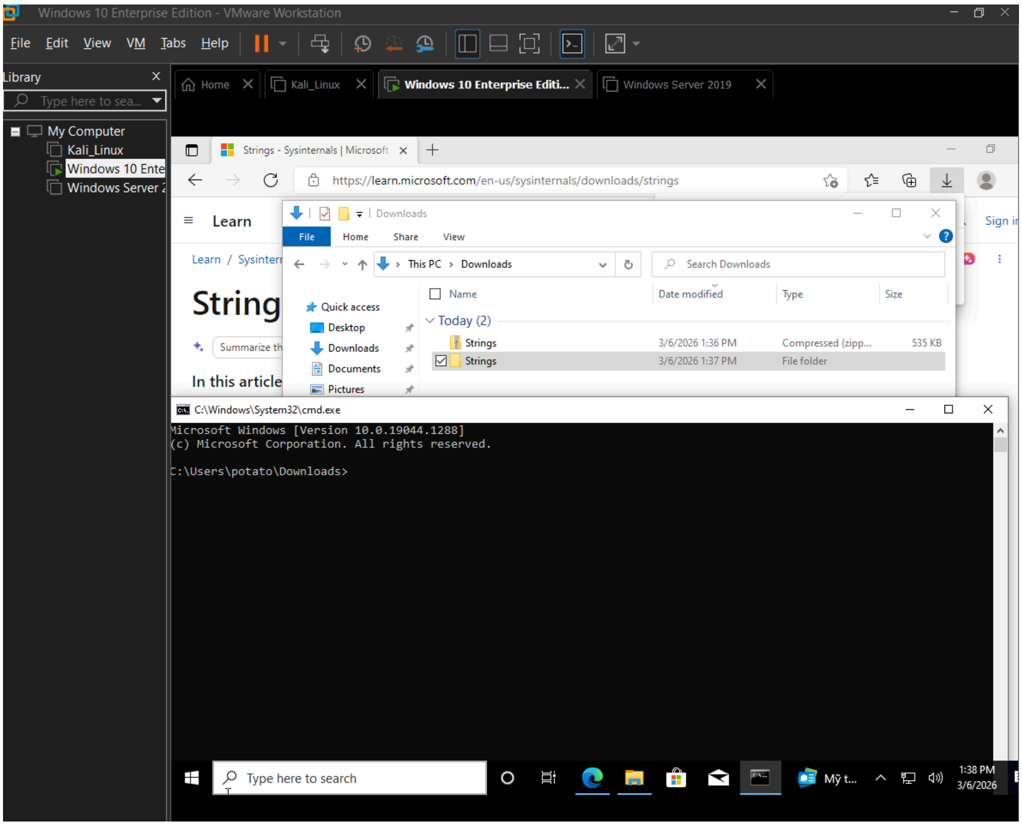
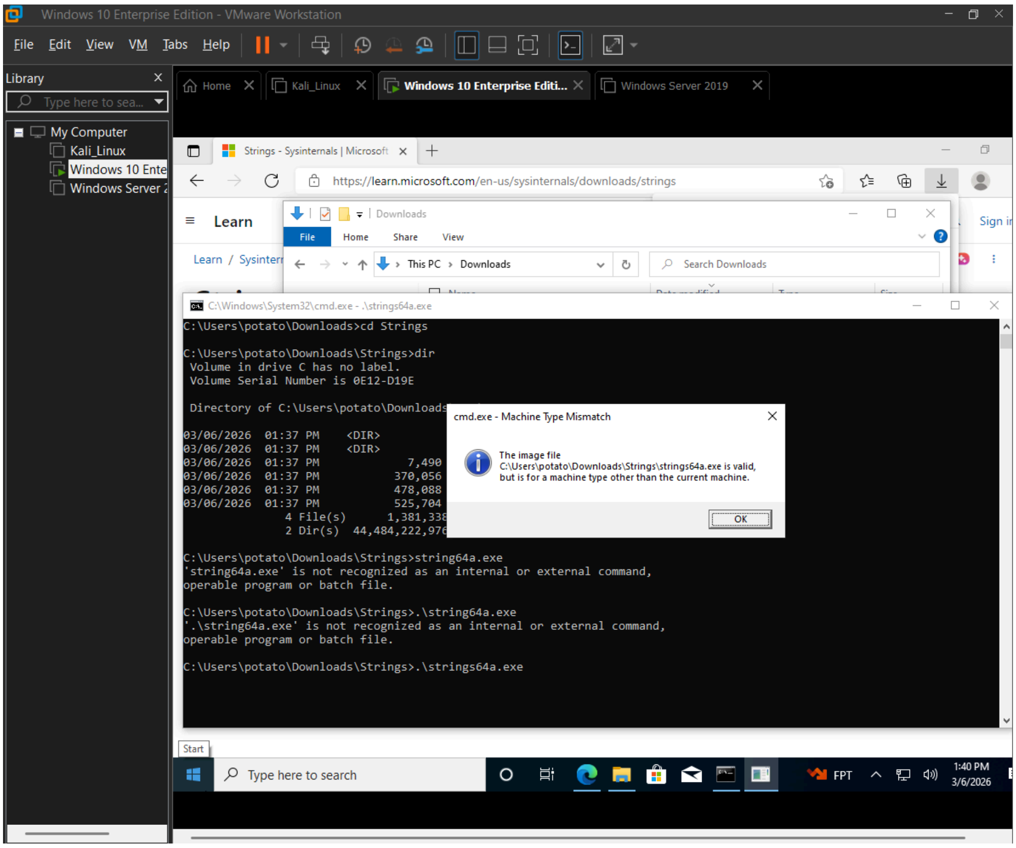
Đoàn Việt Đức - 23020599

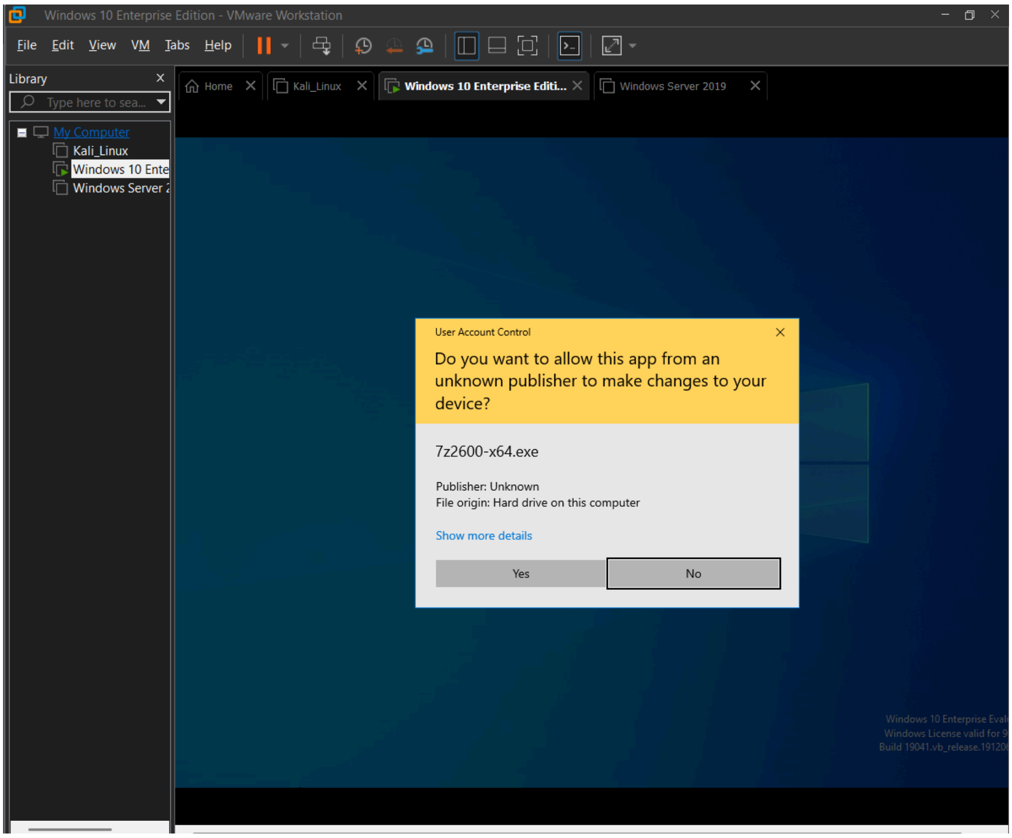
Lab Exercise 15.01: Strings

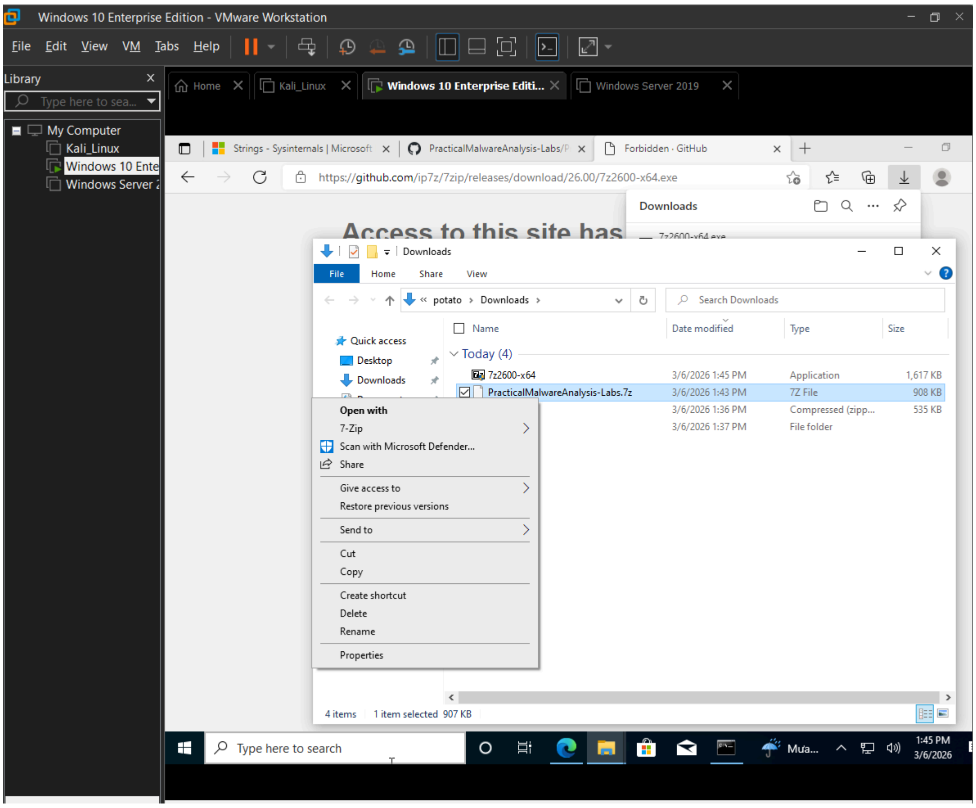
1a-1e

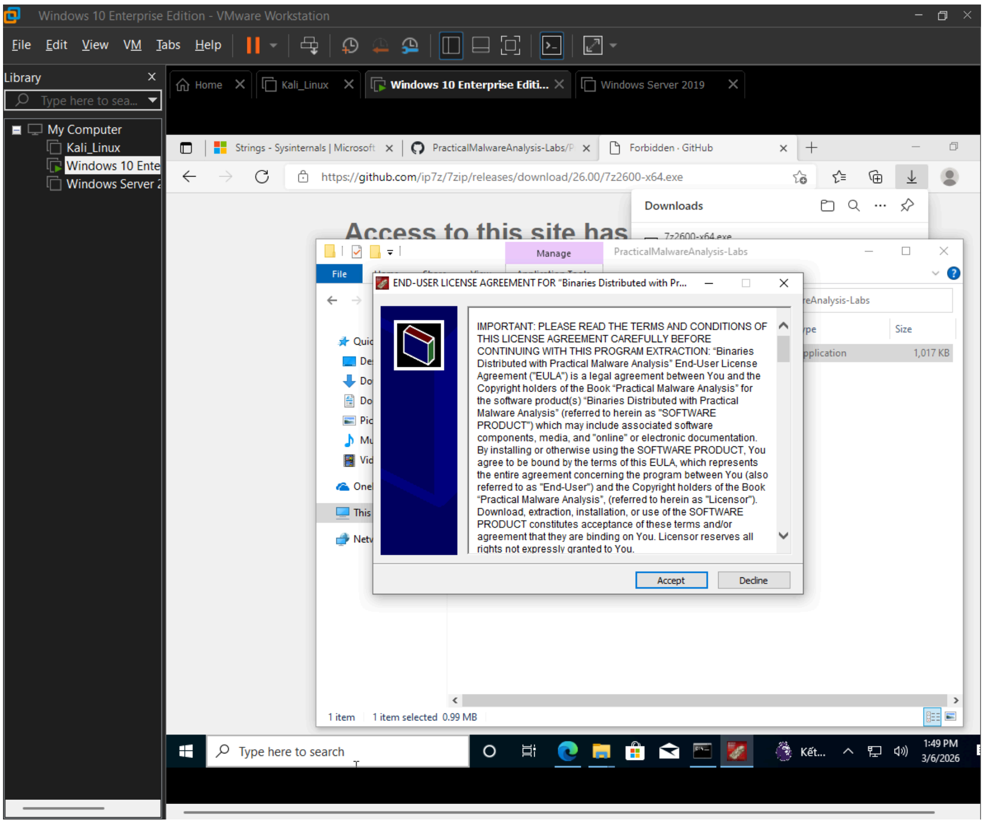



Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Windows 10 Ente
Windows Server 2
Home
Kali_Linux
Windows 10 Enterprise Editi...
Windows Server 2019
Strings - Sysinternals | Microsoft
https://learn.microsoft.com/en-us/sysinternals/downloads/strings
Learn
Downloads
File Home Share View
This PC > Downloads
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Name Date modified Type Size
Today (2)
Strings 3/6/2026 1:36 PM Compressed (zipp... 535 KB
Strings 3/6/2026 1:37 PM File folder
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19044.1288]
(c) Microsoft Corporation. All rights reserved.
C:\Users\potato\Downloads>
Type here to search
1:38 PM
3/6/2026


Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Windows 10 Ente
Windows Server 2
Home
Kali_Linux
Windows 10 Enterprise Editi...
Windows Server 2019
Strings - Sysinternals | Microsoft
https://learn.microsoft.com/en-us/sysinternals/downloads/strings
Learn
Downloads
File Home Share View
This PC > Downloads
Search Downloads
C:\Windows\System32\cmd.exe - .\strings64a.exe
C:\Users\potato\Downloads>cd Strings
C:\Users\potato\Downloads\Strings>dir
Volume in drive C has no label.
Volume Serial Number is 0E12-D19E
Directory of C:\Users\potato\Downloads
03/06/2026 01:37 PM <DIR>
03/06/2026 01:37 PM <DIR>
03/06/2026 01:37 PM 7,490
03/06/2026 01:37 PM 370,056
03/06/2026 01:37 PM 478,088
03/06/2026 01:37 PM 525,704
4 File(s) 1,381,338
2 Dir(s) 44,484,222,976
C:\Users\potato\Downloads\Strings>string64a.exe
'string64a.exe' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\potato\Downloads\Strings>.\string64a.exe
'.\string64a.exe' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\potato\Downloads\Strings>.\strings64a.exe
cmd.exe - Machine Type Mismatch
The image file
C:\Users\potato\Downloads\Strings\strings64a.exe is valid,
but is for a machine type other than the current machine.
OK
Start
Type here to search
FPT
1:40 PM
3/6/2026


Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Home
Windows Server 2019
User Account Control
Do you want to allow this app from an unknown publisher to make changes to your device?
7z2600-x64.exe
Publisher: Unknown
File origin: Hard drive on this computer
Show more details
Yes
No


Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Windows Server 2019
Forbidden - GitHub
https://github.com/ip7z/7zip/releases/download/26.00/7z2600-x64.exe
Downloads
File
Home
Share
View
potato
Search Downloads
Quick access
Desktop
Name
Date modified
Type
Size
Today (4)
7z2600-x64
3/6/2026 1:45 PM
Application
1,617 KB
PracticalMalwareAnalysis-Labs.7z
3/6/2026 1:43 PM
7Z File
908 KB
3/6/2026 1:36 PM
Compressed (zipp...
535 KB
3/6/2026 1:37 PM
File folder
Open with
7-Zip
Scan with Microsoft Defender...
Share
Give access to
Restore previous versions
Send to
Cut
Copy
Create shortcut
Delete
Rename
Properties
4 items
1 item selected 907 KB
Type here to search
1:45 PM
3/6/2026


Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Windows 10 Ente
Windows Server
Home
Kali_Linux
Windows 10 Enterprise Editi...
Windows Server 2019
Strings - Sysinternals | Microsoft
PracticalMalwareAnalysis-Labs/
Forbidden · GitHub
https://github.com/ip7z/7zip/releases/download/26.00/7z2600-x64.exe
Downloads
Access to this site has
Manage
PracticalMalwareAnalysis-Labs
END-USER LICENSE AGREEMENT FOR "Binaries Distributed with Pr...
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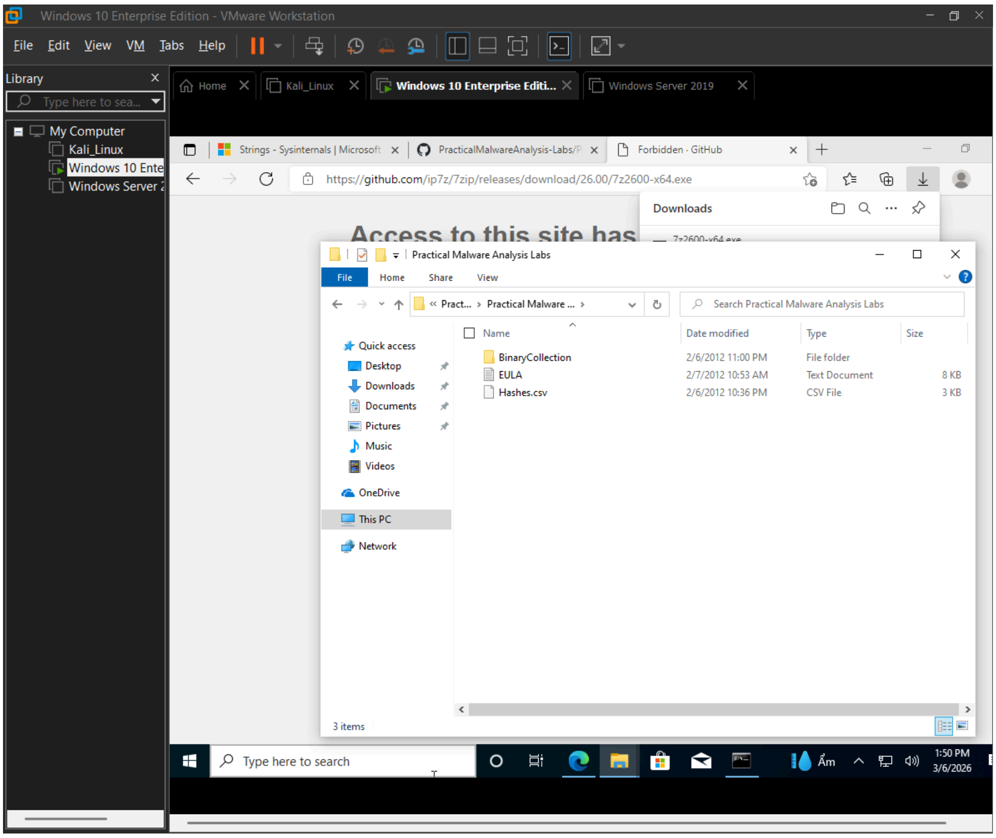
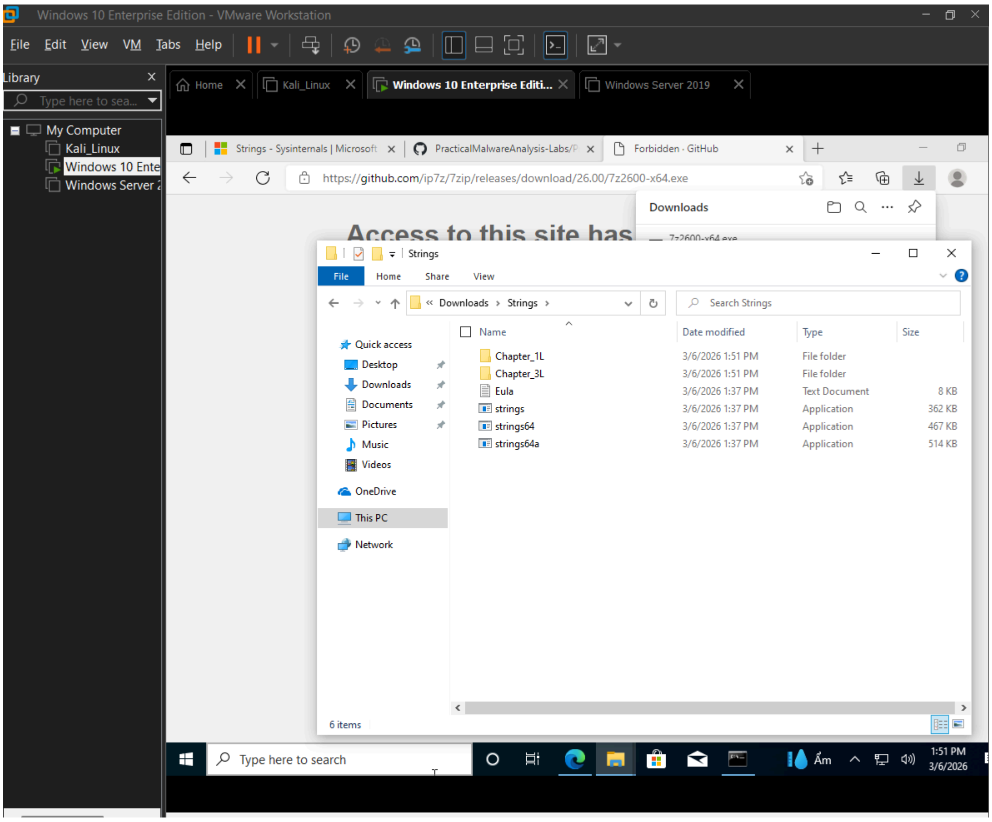
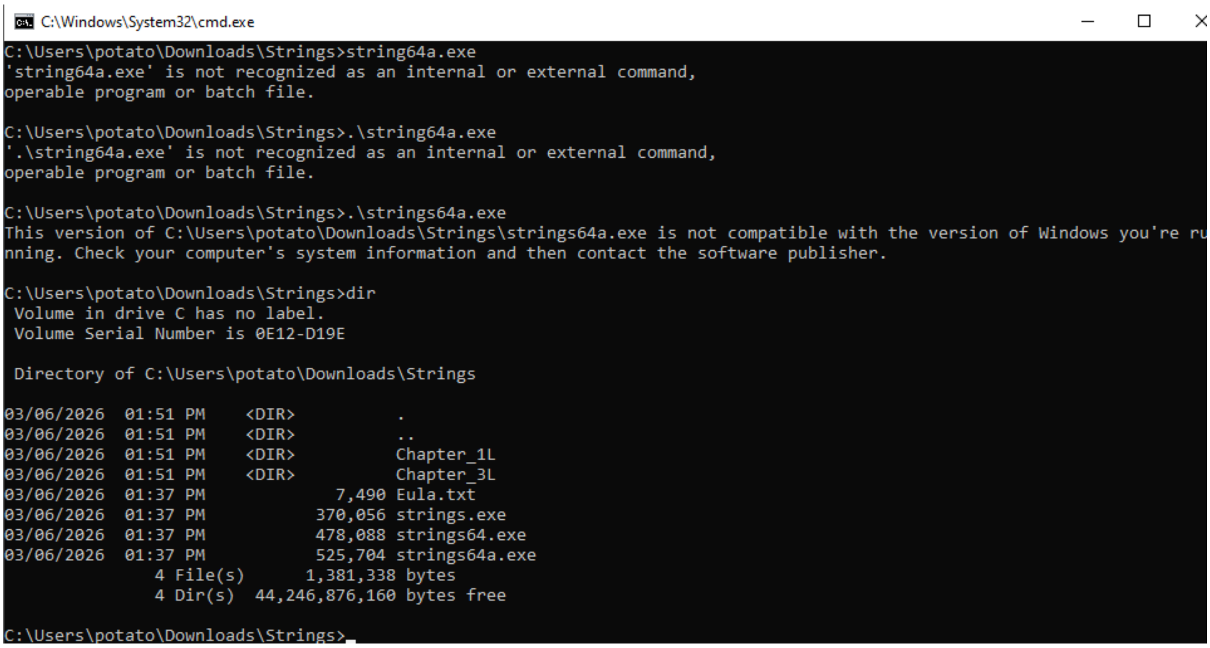
Accept
Decline
Size
1,017 KB
1 item
1 item selected 0.99 MB
Type here to search
1:49 PM
3/6/2026


Windows 10 Enterprise Edition - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Kali_Linux
Windows 10 Ente
Windows Server
Home
Kali_Linux
Windows 10 Enterprise Editi...
Windows Server 2019
Strings - Sysinternals | Microsoft
PracticalMalwareAnalysis-Labs/
Forbidden · GitHub
https://github.com/ip7z/7zip/releases/download/26.00/7z2600-x64.exe
Downloads
Access to this site has
Practical Malware Analysis Labs
File Home Share View
« Pract... › Practical Malware ... ›
Search Practical Malware Analysis Labs
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
BinaryCollection 2/6/2012 11:00 PM File folder
EULA 2/7/2012 10:53 AM Text Document 8 KB
Hashes.csv 2/6/2012 10:36 PM CSV File 3 KB
3 items
Type here to search
1:50 PM
3/6/2026





```
C:\Windows\System32\cmd.exe
C:\Users\potato\Downloads\Strings>string64a.exe
'string64a.exe' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\potato\Downloads\Strings>.\string64a.exe
'.\string64a.exe' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\potato\Downloads\Strings>.\strings64a.exe
This version of C:\Users\potato\Downloads\Strings\strings64a.exe is not compatible with the version of Windows you're ru
nning. Check your computer's system information and then contact the software publisher.

C:\Users\potato\Downloads\Strings>dir
 Volume in drive C has no label.
 Volume Serial Number is 0E12-D19E

 Directory of C:\Users\potato\Downloads\Strings

03/06/2026  01:51 PM    <DIR>          .
03/06/2026  01:51 PM    <DIR>          ..
03/06/2026  01:51 PM    <DIR>          Chapter_1L
03/06/2026  01:51 PM    <DIR>          Chapter_3L
03/06/2026  01:37 PM             7,490 Eula.txt
03/06/2026  01:37 PM           370,056 strings.exe
03/06/2026  01:37 PM           478,088 strings64.exe
03/06/2026  01:37 PM           525,704 strings64a.exe
               4 File(s)      1,381,338 bytes
               4 Dir(s)  44,246,876,160 bytes free

C:\Users\potato\Downloads\Strings>
```

```
C:\Users\potato\Downloads\Strings>strings64 Lab01-01.dll | more


Strings v2.54 - Search for ANSI and Unicode strings in binary images.
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Sysinternals - www.sysinternals.com

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*supplements,
*Internet - based services,
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ll other rights.Unless applicable law gives you more rights despite this limitation, you may use the software only as ex
```

```
C:\Users\potato\Downloads\Strings\Chapter_1L>strings64 Lab01-01.dll

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
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Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.reloc
SUV
h8`
h8`
L$xQh
h(`
RVf
D$"
-(
IQh `
|$4
D$$
L$4PQj
D$\D
_^]
u?h
%d`
Y^j
```

```
C:\Windows\System32\cmd.exe

C:\Users\potato\Downloads\Strings\Chapter_1L>strings64 -n 4 Lab01-01.dll | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
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Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.reloc
L$xQh
IQh `
L$4PQj
D$\D
NWVS
u7WPS
u&WVS
_^[]
CloseHandle
Sleep
CreateProcessA
CreateMutexA
OpenMutexA
KERNEL32.dll
WS2_32.dll
strncmp
```

```
C:\Windows\System32\cmd.exe
@.data
.reloc
L$xQh
IQh `
L$4PQj
D$\D
NWVS
u7WPS
u&WVS
_^[]
CloseHandle
Sleep
CreateProcessA
CreateMutexA
OpenMutexA
KERNEL32.dll
WS2_32.dll
strncmp
MSVCRT.dll
free
_initterm
malloc
_adjust_fdiv
exec
sleep
hello
127.26.152.13
SADFHUHF
/0I0[0h0p0
-- More  --
```

```
C:\Users\potato\Downloads\Strings\Chapter_1L>strings64 -n 4 -o lab01-01.dll | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
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Sysinternals - www.sysinternals.com

0077:!This program cannot be run in DOS mode.
0192:Rich
0472:.text
0511:`.rdata
0551:@.data
0592:.reloc
4213:L$xQh
4345:IQh `
4489:L$4PQj
4520:D$\D
4929:NWVS
4948:u7WPS
4965:u&WVS
5008:_^[]
8458:CloseHandle
8472:Sleep
```

```
lab01-01 - Notepad
File Edit Format View Help
0077:!This program cannot be run in DOS mode.
0192:Rich
0472:.text
0511:`.rdata
0551:@.data
0592:.reloc
4213:L$xQh
4345:IQh `
4489:L$4PQj
4520:D$\D
4929:NWVS
4948:u7WPS
4965:u&WVS
5008:_^[]
8458:CloseHandle
8472:Sleep
8480:CreateProcessA
8498:CreateMutexA
8514:OpenMutexA
8526:KERNEL32.dll
8540:WS2_32.dll
8554:strncmp
8562:MSVCRT.dll
8576:free
```

```
lab01-01 - Notepad
File Edit Format View Help
8458:CloseHandle
8472:Sleep
8480:CreateProcessA
8498:CreateMutexA
8514:OpenMutexA
8526:KERNEL32.dll
8540:WS2_32.dll
8554:strncmp
8562:MSVCRT.dll
8576:free
8584:_initterm
8596:malloc
8606:_adjust_fdiv
155664:exec
155672:sleep
155680:hello
155688:127.26.152.13
155704:SADFHUHF
159752:/0I0[0h0p0
159785:141G1[1l1
159801:1Y2a2g2r2
159835:3!3}3
```

2a-2d

```
C:\Users\potato\Downloads\Strings\Chapter_1L>strings64 -n 4 Lab01-01.exe

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!This program cannot be run in DOS mode.
Richm
.text
`.rdata
@.data
UVWj
@jjj
@jjj
ugh 0@
_^][
SUVW
h00@
_^][
SUVW
@jjj
@jjj
h|0@
D$Pj
l$\u
S$QWR
FxRVP
D$$3
D$8R
t$<f
T$PR
hL0@
h|0@
hD0@
_^]3
hp @
 SVW
%3 @
%D @
%\ @
```

```
Select C:\Windows\System32\cmd.exe
%D @
%\ @
%` @
CloseHandle
UnmapViewOfFile
IsBadReadPtr
MapViewOfFile
CreateFileMappingA
CreateFileA
FindClose
FindNextFileA
FindFirstFileA
CopyFileA
KERNEL32.dll
malloc
exit
MSVCRT.dll
_exit
_XcptFilter
__p___initenv
__getmainargs
_initterm
__setusermatherr
_adjust_fdiv
__p__commode
__p__fmode
__set_app_type
_except_handler3
_controlfp
_stricmp
kerne132.dll
kernel32.dll
.exe
C:\*
C:\windows\system32\kerne132.dll
Kernel32.
Lab01-01.dll
C:\Windows\System32\Kernel32.dll
WARNING_THIS_WILL_DESTROY_YOUR_MACHINE
```

Functions, visible as strings, that deal with finding, opening, and manipulating files:

- FindFirstFileA — Starts directory/file enumeration (finding files).
- FindNextFileA — Continues enumeration of the next file.
- FindClose — Closes a file search handle.
- CreateFileA — Opens or creates a file.
- CopyFileA — Copies an existing file to a new location.
- CreateFileMappingA — Creates a file mapping object (for memory-mapped file I/O).
- MapViewOfFile — Maps a view of a file into the address space (manipulating file contents in memory).
- UnmapViewOfFile — Unmaps a previously mapped file view.
- CloseHandle — Closes file (or other) handles after operations.

String indicating a type of file that this malware might be searching for:

- **C:***
- This wildcard string (seen in Lab01-01.exe strings) indicates the malware is likely enumerating/searching all files (or directories) starting from the root of the C: drive. It is probably used to locate the Windows installation folder (e.g., to find system32) or to scan broadly for executables to infect/modify.

String that appears to be spoofing an actual file by altering one character:

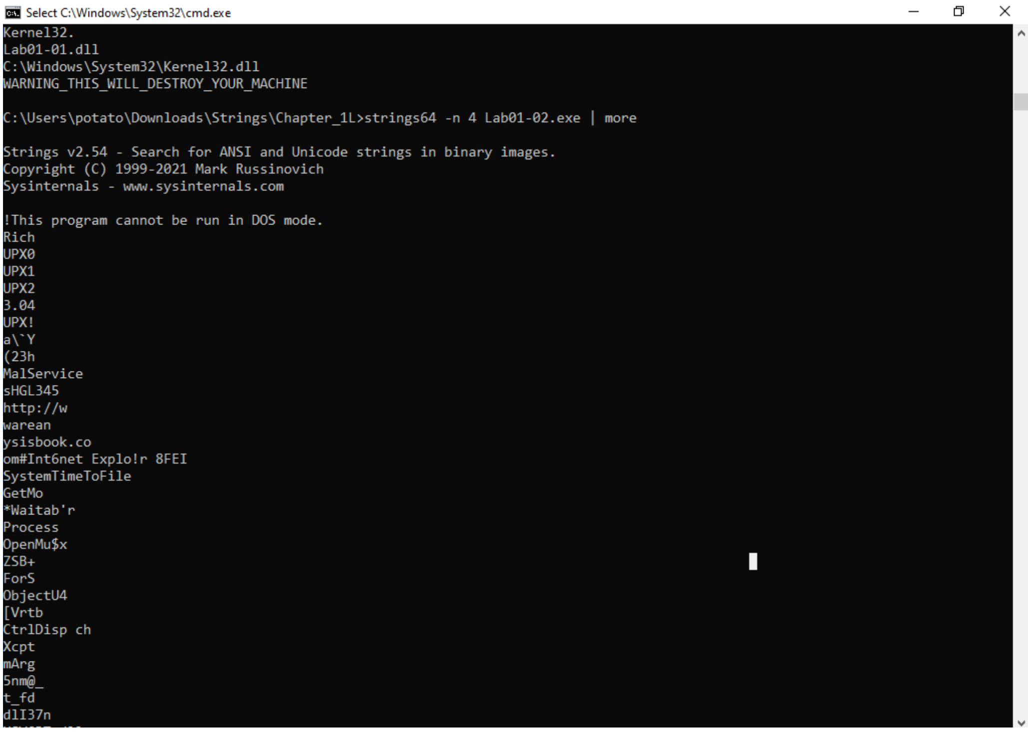
kerne132.dll (with the number 1 instead of a lowercase l — visually very similar in most fonts: kerne132.dll vs kernel32.dll).

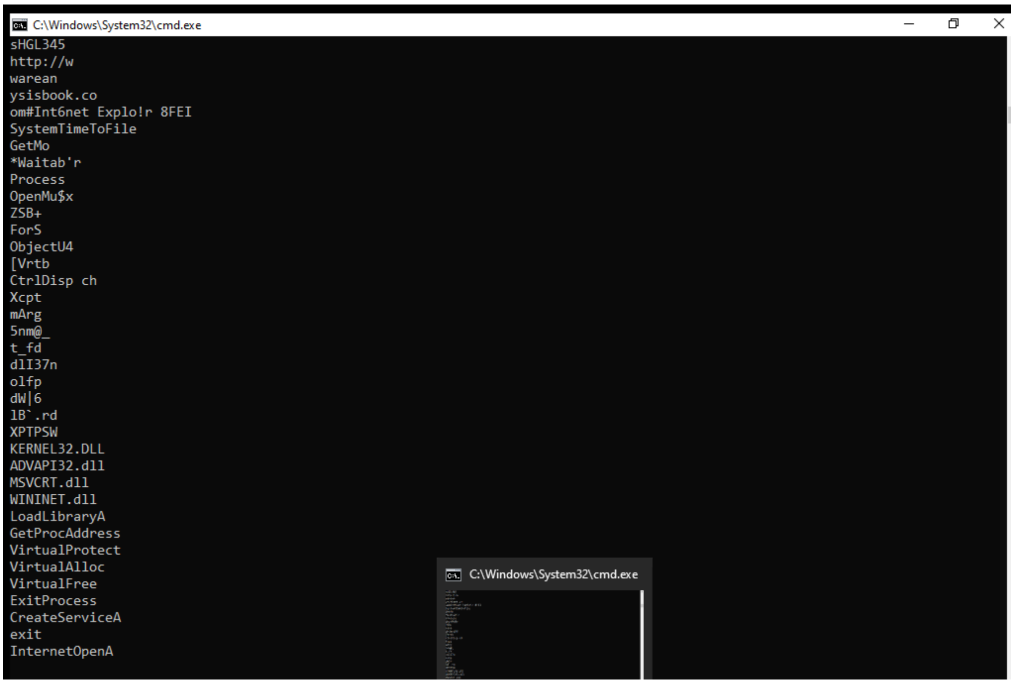
Two other problems with that string:

- The path uses lowercase "c:\windows\system32\kerne132.dll" (all lowercase drive letter and folder names), whereas the legitimate Windows path is typically "C:\Windows\System32\kernel32.dll" (title case for "Windows" and "System32").
- Inconsistent/inaccurate casing in the filename itself (e.g., starts with lowercase "k" in some references, while the real one is usually referenced as "Kernel32.dll" in title case in system contexts).

The legitimate string it is trying to masquerade as (also visible in the strings output of the EXE) is kernel32.dll (or the full path C:\Windows\System32\Kernel32.dll / kernel32.dll).

3a-3c





DLL and function hinting to network communication:

The binary references WININET.dll and the function InternetOpenA. WININET.dll is a Windows library for internet protocols (e.g., HTTP), and InternetOpenA initializes a WinINet session for network operations like connecting to URLs or downloading data.

Explanation of UPX:

UPX (Ultimate Packer for eXecutables) is a free, open-source, portable, and high-performance executable compressor that supports multiple formats, typically reducing file sizes by 50-70% through compression while allowing quick decompression with no significant memory overhead. **It works by adding a stub to the executable that unpacks the compressed code at runtime**. UPX is significant for analyzing this binary because the presence of strings like "UPX0", "UPX1", and "UPX!" (section names) confirms it's packed with UPX, a common technique used by malware authors to obfuscate code and evade static analysis or detection tools.

Hypothesis

Lab01-02.exe appears to be a UPX-packed malware sample designed for persistence and network activity. The packing hides most details, but visible artifacts suggest it installs a Windows service named "MalService" (using APIs like CreateServiceA and OpenMutexA for single-instance checks). It likely establishes an internet connection via WININET.dll functions (e.g., InternetOpenA) to communicate with the reconstructed URL "http://www.practicalmalwareanalysis.com", possibly for command-and-control (C2) or downloading additional payloads. The fake warning "WARNING_THIS_WILL_DESTROY_YOUR_MACHINE" is to deter analysis.

4a-4b

```
C:\Users\potato\Downloads\Strings\Chapter_1L>strings64 -n 4 lab01-03.exe | more

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!Windows Program
`.rdata
@.data
KERNEL32.dll
LoadLibraryA
GetProcAddress
":Ll
3Bt>O
VQ(8
2]<,M
P@M^
 S>VW
AQ=h
I*G9>
e%nN
ole32.vd
Init
FoCr
U!!C
}OLEAUTLA
IMSVCRTT"b
_getmas
yrcs
|P2r3Us
p|vuy
fmod
xF*l
```
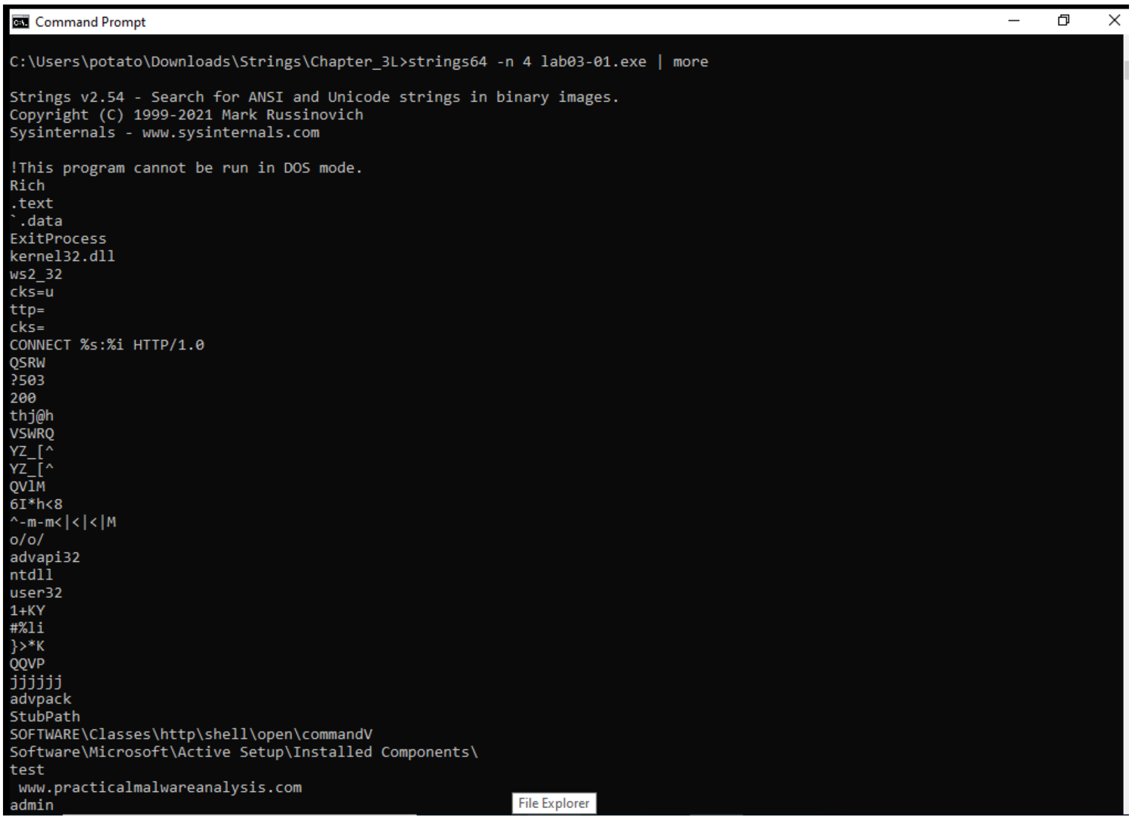
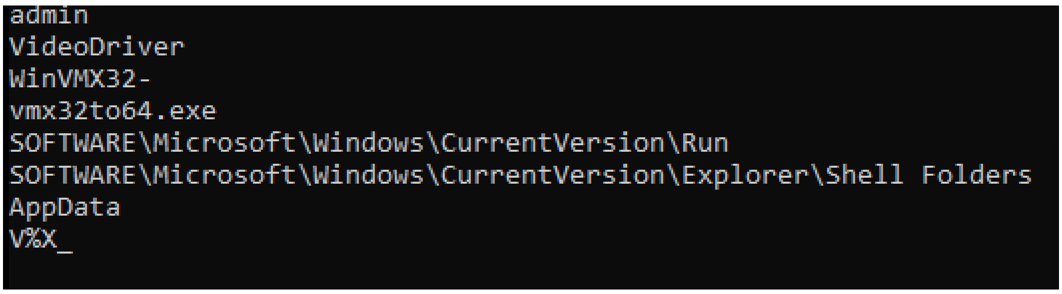

There are fewer meaningful strings compared to previous .exe files. This could indicate that the malware is highly obfuscated, which makes it hard for static analysis.

5a-5e





The following strings represent registry keys or paths, likely used for persistence or configuration:

- SOFTWARE\Classes\http\shell\open\commandV

- Software\Microsoft\Active Setup\Installed Components\
- SOFTWARE\Microsoft\Windows\CurrentVersion\Run
- SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders

These point to common persistence mechanisms, such as installing components or running executables on startup.

String that is an FQDN

The fully qualified domain name (FQDN) is www.practicalmalwareanalysis.com. This likely serves as a command-and-control (C2) server for exfiltration, downloads, or beaconing.

Strings related to video

The string VideoDriver appears related to video functionality. It may be used as a registry value or key name to masquerade as a legitimate video driver component for persistence or evasion.

String tied to a username for adversary login

The string **admin** appears to be a hardcoded credential (likely a username or password) that an adversary could use to authenticate to the malware if it provides a server or backdoor service.

Hypothesis

Lab03-01.exe seems to be a packed backdoor or trojan with network capabilities. It likely unpacks at runtime (evidenced by minimal static imports like ExitProcess from kernel32.dll, but strings hinting at dynamic loading of libraries like advapi32, ntdll, user32, and possibly ws2_32 for networking). For persistence, it copies itself to vmx32to64.exe in the system directory and sets registry entries (e.g., under Run key) to execute on startup, using VideoDriver as a value to blend in. A mutex like WinVMX32 prevents multiple instances. It establishes outbound connections to www.practicalmalwareanalysis.com using HTTP CONNECT proxies (via strings like "CONNECT %s:%i HTTP/1.0"), potentially for C2 communication, data exfiltration, or payload downloads. The "admin" string suggests authentication for remote access, making it a basic remote access tool (RAT)

6a-6f

Select Command Prompt

```
C:\Users\potato\Downloads\Strings\Chapter_3L>strings64 -n 4 lab03-02.dll | more

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!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.reloc
QQSUVW3
Hu4S
PSUV
j@SU
D$ 3
|$%Y
D$$SPh
t%HHt
D$ P
D$$h
D$ P
PhDa
SSSSh
tNSh
Qh4a
SSSSP
_WSP
QSSSSSS
F;5p
SVW3
PhDa
SSSj
Ph~f
SVWjJ3
SPSS
tFSh
VSh0
SSSPS
u(hta
uZh`a
PSSj
```
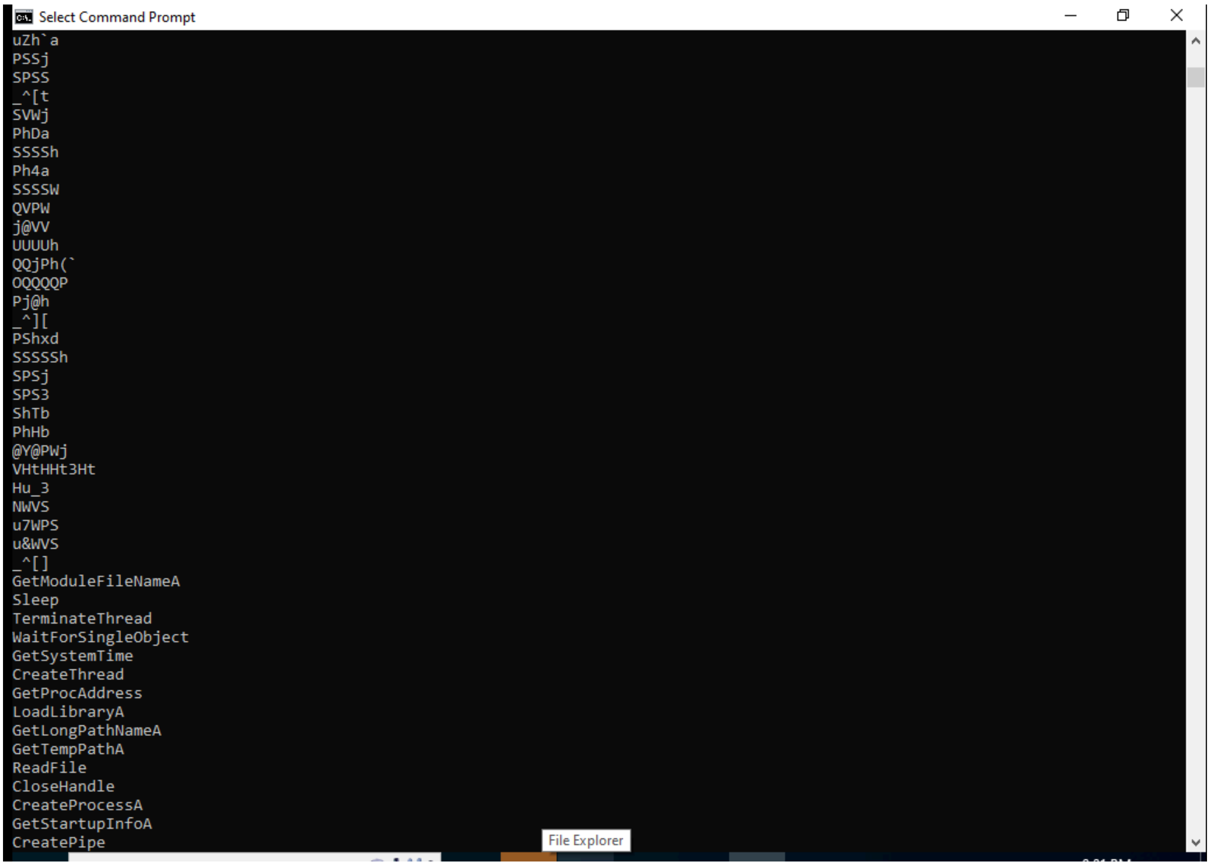
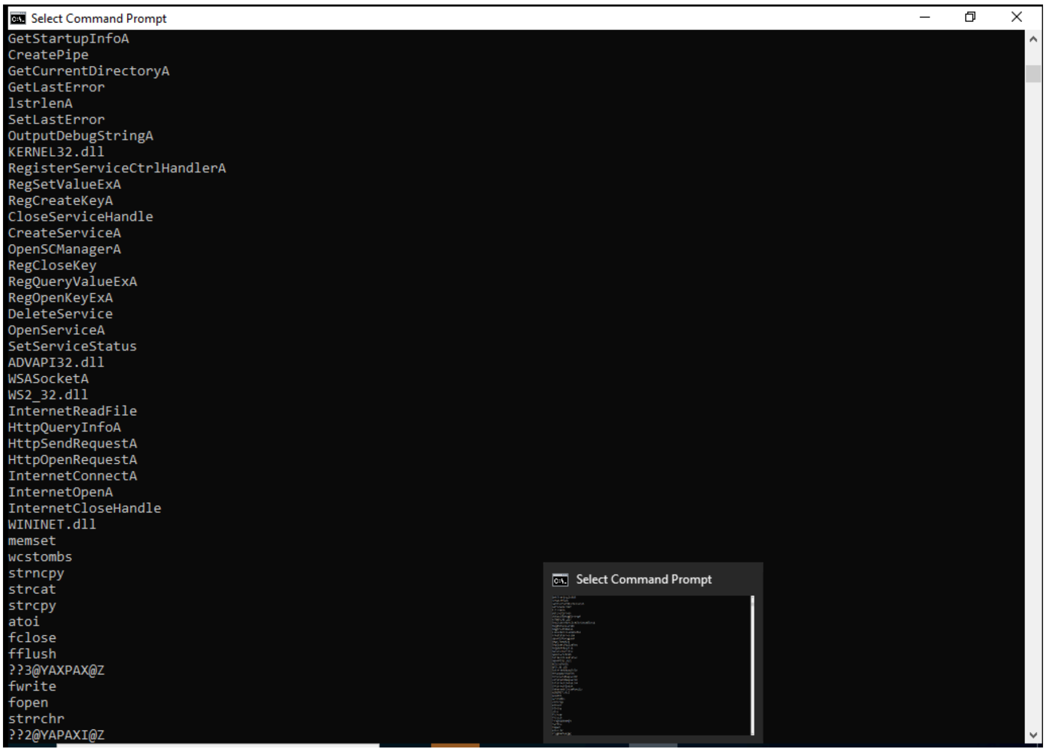
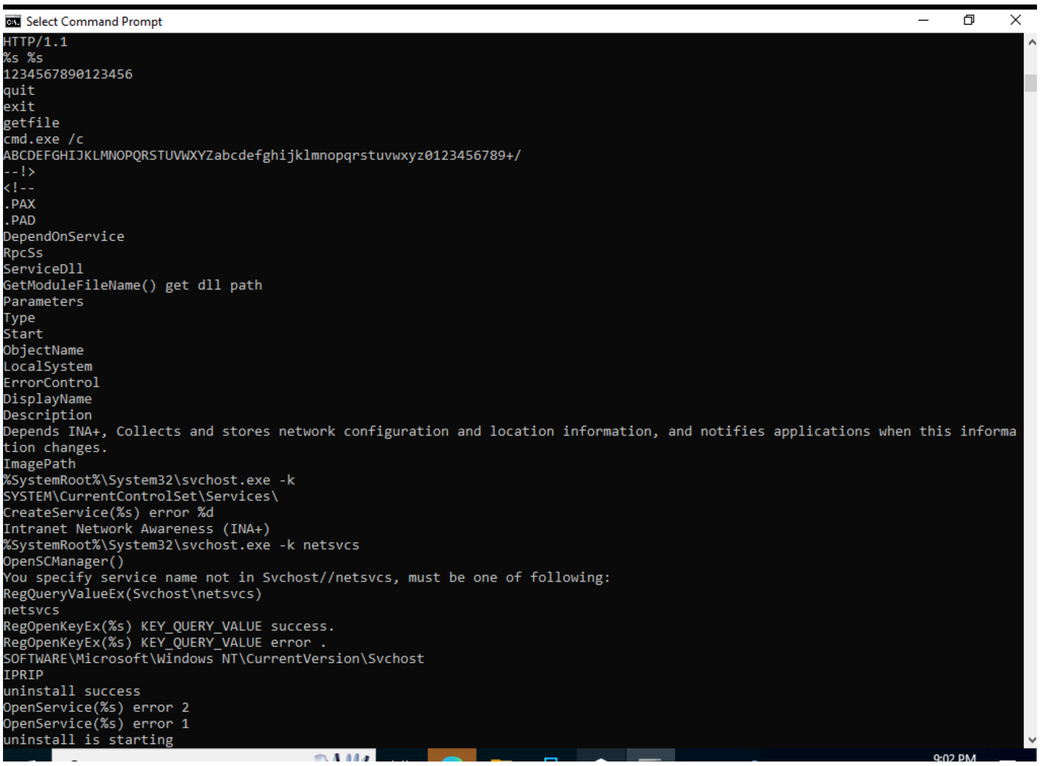

File Explorer



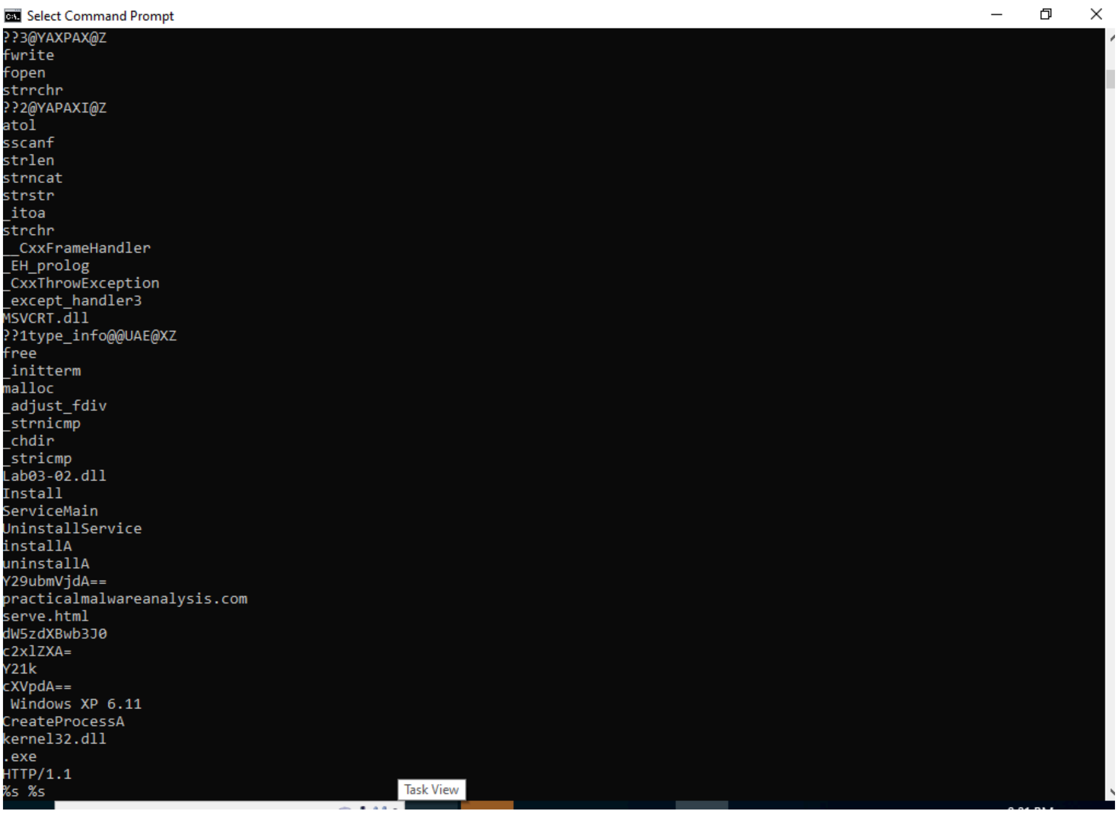
```
Select Command Prompt
uZh`a
PSSj
SPSS
_^[t
SVWj
PhDa
SSSSh
Ph4a
SSSSW
QVPW
j@VV
UUUUh
QQjPh(`
OQQQQP
Pj@h
_^][
PShxd
SSSSSh
SPSj
SPS3
ShTb
PhHb
@Y@PWj
VHtHHt3Ht
Hu_3
NWVS
u7WPS
u&WVS
_^[]
GetModuleFileNameA
Sleep
TerminateThread
WaitForSingleObject
GetSystemTime
CreateThread
GetProcAddress
LoadLibraryA
GetLongPathNameA
GetTempPathA
ReadFile
CloseHandle
CreateProcessA
GetStartupInfoA
CreatePipe
File Explorer
```



```
Select Command Prompt
GetStartupInfoA
CreatePipe
GetCurrentDirectoryA
GetLastError
lstrlenA
SetLastError
OutputDebugStringA
KERNEL32.dll
RegisterServiceCtrlHandlerA
RegSetValueExA
RegCreateKeyA
CloseServiceHandle
CreateServiceA
OpenSCManagerA
RegCloseKey
RegQueryValueExA
RegOpenKeyExA
DeleteService
OpenServiceA
SetServiceStatus
ADVAPI32.dll
WSASocketA
WS2_32.dll
InternetReadFile
HttpQueryInfoA
HttpSendRequestA
HttpOpenRequestA
InternetConnectA
InternetOpenA
InternetCloseHandle
WININET.dll
memset
wcstombs
strncpy
strcat
strcpy
atoi
fclose
fflush
??3@YAXPAX@Z
fwrite
fopen
strrchr
??2@YAPAXI@Z
Select Command Prompt
```



```
HTTP/1.1
%s %s
1234567890123456
quit
exit
getfile
cmd.exe /c
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/
--!>
<!--
.PAX
.PAD
DependOnService
RpcSs
ServiceDll
GetModuleFileName() get dll path
Parameters
Type
Start
ObjectName
LocalSystem
ErrorControl
DisplayName
Description
Depends INA+, Collects and stores network configuration and location information, and notifies applications when this informa
tion changes.
ImagePath
%SystemRoot%\System32\svchost.exe -k
SYSTEM\CurrentControlSet\Services\
CreateService(%s) error %d
Intranet Network Awareness (INA+)
%SystemRoot%\System32\svchost.exe -k netsvcs
OpenSCManager()
You specify service name not in Svchost//netsvcs, must be one of following:
RegQueryValueEx(Svchost\netsvcs)
netsvcs
RegOpenKeyEx(%s) KEY_QUERY_VALUE success.
RegOpenKeyEx(%s) KEY_QUERY_VALUE error .
SOFTWARE\Microsoft\Windows NT\CurrentVersion\Svchost
IPRIP
uninstall success
OpenService(%s) error 2
OpenService(%s) error 1
uninstall is starting
```



```
??3@YAXPAX@Z
fwrite
fopen
strrchr
??2@YAPAXI@Z
atol
sscanf
strlen
strncat
strstr
_itoa
strchr
__CxxFrameHandler
_EH_prolog
_CxxThrowException
_except_handler3
MSVCRT.dll
??1type_info@@UAE@XZ
free
_initterm
malloc
_adjust_fdiv
_strnicmp
_chdir
_stricmp
Lab03-02.dll
Install
ServiceMain
UninstallService
installA
uninstallA
Y29ubmVjdA==
practicalmalwareanalysis.com
serve.html
dW5zdXBwb3J0
c2xlZXA=
Y21k
cXVpdA==
 Windows XP 6.11
CreateProcessA
kernel32.dll
.exe
HTTP/1.1
%s %s
```

The following strings are Windows API functions or custom exports related to service manipulation:

- CreateServiceA: Creates a new service in the Service Control Manager (SCM).

- OpenServiceA: Opens an existing service for control.
- CloseServiceHandle: Closes a handle to a service or SCM.
- RegisterServiceCtrlHandlerA: Registers a handler for service control requests.
- SetServiceStatus: Updates the status of the service (appears as "setServiceStatus" in strings, likely a wrapper or typo for the API).
- OpenSCManagerA: Opens the SCM database.
- StartServiceCtrlDispatcherA (implied from context, though not directly visible; used to dispatch service threads).
- DeleteService: Deletes a service (from related uninstall strings). Custom exports like ServiceMain, Install, UninstallService, InstallA, and uninstallA indicate the DLL's entry points for installing/uninstalling as a service.

Strings that are functions manipulating the registry

- RegSetValueExA: Sets a value in a registry key.
- RegCreateKeyA: Creates a new registry key.
- RegCloseKey: Closes a registry key handle.
- RegOpenKeyEx: Opens an existing registry key (with access flags like KEY_QUERY_VALUE).
- RegQueryValueEx: Queries a value from a registry key. These are used for persistence, such as adding the service to HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Svchost\netsvcs.

Strings related to networking

- WSAStartup: Initializes Winsock for socket operations.
- InternetReadFile: Reads data from an internet handle.
- HttpQueryInfoA: Queries HTTP response headers.
- HttpSendRequestA: Sends an HTTP request.
- HttpOpenRequestA: Creates an HTTP request handle.
- InternetConnectA: Connects to a server.
- InternetOpenA: Initializes a WinINet session.
- InternetCloseHandle: Closes an internet handle.
- WININET.dll and WS2_32.dll: Libraries for HTTP and socket networking.
- HTTP/1.1 %s %s: HTTP request/response format string.
- practicalmalwareanalysis.com and serve.html: Likely a C2 domain and endpoint.

Option in the string that contains "cmd" and why it's suspicious

The string is "cmd.exe /c". The "/c" option instructs cmd.exe to execute the command string that follows it and then terminate. This is suspicious because malware frequently uses "cmd.exe /c" to run arbitrary system commands (e.g., for downloading payloads, reconnaissance, or lateral movement) in a non-interactive, hidden manner without leaving a persistent shell window

String with uppercase letters, lowercase letters, numbers, and strings

The string "ABCDEFG HIJKLMNOPQR STUV WXYZ abcdefgh ijklmnop qrstu vwx yz0123456789+/" (note: spaces may be artifacts from string extraction) is the standard Base64 character set (A-Z, a-z, 0-9, +, /). This is suspicious because it indicates the binary implements Base64 encoding/decoding, commonly used in malware to obfuscate command-and-control (C2) data, payloads, or exfiltrated information. Supporting evidence includes nearby encoded strings like "V29ubmVjdA==" (decodes to "connect"), "c2xlZXA=" (decodes to "sleep"), and "Y21k" (decodes to "cmd"), suggesting command parsing.

Hypothesis

Lab03-02.dll is a malicious DLL designed as a backdoor service. It exports functions like InstallA and uninstallA for self-installation via rundll32.exe (e.g., rundll32.exe Lab03-02.dll,installA), creating a fake service named IPRIP (masquerading as "Intranet Network Awareness (INA+)") that runs under svchost.exe -k netsvcs for persistence. It modifies the registry (e.g., under Svchost\netsvcs) to ensure loading on boot. Once running, it establishes HTTP connections to practicalmalwareanalysis.com/serve.html for C2 communication, using Base64 to encode/decode commands such as "sleep" (delay execution), "cmd" (run shell commands via cmd.exe /c), "quit/exit" (terminate), "getfile" (download), and potentially upload. Additional capabilities include thread creation, process spawning, file manipulation (e.g., GetTempPathA, DeleteFileA), and error handling. Overall, it functions as a persistent remote access trojan (RAT) with command execution and data transfer features.

7a-7b

```
C:\Users\potato\Downloads\Strings\Chapter_3L>strings64 -n 4 lab03-03.exe | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
.rsrc
jjjj
h@P@
hXP@
hdP@
hlP@
Ph0P@
GIt#
YYh P@
t9UW
?=t"U
QQS3
PSSW
8"uD
8"uF@
8"u,
@@f9
@@f9
=|@@
SS@SSPVSS
t#SSUP
t$$VSS
_^][YY
DSUVWh
_^][
SVWUj
]_^[
t.;t$$t(
VC20XC00U
SVWU
tEVU
```

```
VC20XC00U
SVWU
tEVU
t3x<
]_^[
hxC@
58T@
90tr
%(T@
Wj@Y3
t7SW

@AA;
%tT@
%xT@
j?I_
ulSj
5lT@
uY;]
pD#U
j #M
j?^;
%tT@
5|T@
VWuBh
5(@@
tzVS
GIt%
t/Ku
h D@
9=`T@
uFWWj
"WWSh
9} u
E WW
tMWWS
t@9}
VSh
h8D@
runtime error
TLOSS error
SING error
```

```
DOMAIN error
R6028
- unable to initialize heap
R6027
- not enough space for lowio initialization
R6026
- not enough space for stdio initialization
R6025
- pure virtual function call
R6024
- not enough space for _onexit/atexit table
R6019
- unable to open console device
R6018
- unexpected heap error
R6017
- unexpected multithread lock error
R6016
- not enough space for thread data
abnormal program termination
R6009
- not enough space for environment
R6008
- not enough space for arguments
R6002
- floating point not loaded
-- More --
```

```
Microsoft Visual C++ Runtime Library
Runtime Error!
Program:
<program name unknown>
GetLastActivePopup
GetActiveWindow
MessageBoxA
user32.dll
CloseHandle
VirtualFree
ReadFile
VirtualAlloc
GetFileSize
CreateFileA
ResumeThread
SetThreadContext
WriteProcessMemory
VirtualAllocEx
GetProcAddress
GetModuleHandleA
ReadProcessMemory
GetThreadContext
CreateProcessA
FreeResource
SizeofResource
LockResource
LoadResource
FindResourceA
GetSystemDirectoryA
Sleep
KERNEL32.dll
GetCommandLineA
GetVersion
ExitProcess
TerminateProcess
GetCurrentProcess
UnhandledExceptionFilter
GetModuleFileNameA
FreeEnvironmentStringsA
FreeEnvironmentStringsW
WideCharToMultiByte
GetEnvironmentStrings
```

Hypothesis

The string outputs show a lot of potentially obfuscated, nonsense strings. There are some runtime errors, mostly out-of-space warnings. This indicates that the .exe file might be trying to execute a buffer overflow attack to insert malicious code into memory region. The list of WinAPI strings listed: **VirtualAllocEx**, **CreateProcessA, …** suggests that the malware is trying to grant execution permission for a specific area inside the memory region, confirming that there might be a buffer overflow attack.

8a-8d

```
C:\Users\potato\Downloads\Strings\Chapter_3L>strings64 -n 4 lab03-04.exe | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
6KRich
.text
`.rdata
@.data
jjjj
SVWh
SVWh
SVWh
jjjjjj
jjjjj
SVWh?
YYh(
SUVW
_^][
QSVW
t7)E
^[_3
Yt-j
tD+E
SVWu
X_[^
HHtpHHtl
Yu!j
^_[3
<8=u
7t'It
SUVW
_^][
DSUVWh
_^][
@u;9
tjVj
SVWu
Y;50
90tr
```

Microsoft Store

Select Command Prompt

```
QQVW
t;j\V
SVW3
^VWP
8=u)
t#PV
_^[]
0SVW
_u@W
PWPSS
PWPSS
9] u
tySS
t-VW
QQSVW3
tUj=
t@9u
uT9}
8<=t
^][_
command.com
COMSPEC
 (8PX
700WP
`h````
ppxxxx
(null)
(null)
__GLOBAL_HEAP_SELECTED
__MSVCRT_HEAP_SELECT
runtime error
TLOSS error
SING error
DOMAIN error
R6028
- unable to initialize heap
R6027
- not enough space for lowio initialization
R6026
- not enough space for stdio initialization
R6025
- pure virtual function call
R6024
```

Select Command Prompt

```
GetLastError
ReadFile
WriteFile
Sleep
GetShortPathNameA
GetModuleFileNameA
CopyFileA
ExpandEnvironmentStringsA
DeleteFileA
KERNEL32.dll
RegQueryValueExA
RegOpenKeyExA
RegSetValueExA
RegCreateKeyExA
RegDeleteValueA
CreateServiceA
CloseServiceHandle
ChangeServiceConfigA
OpenServiceA
OpenSCManagerA
DeleteService
ADVAPI32.dll
ShellExecuteA
SHELL32.dll
WS2_32.dll
ExitProcess
TerminateProcess
GetCurrentProcess
GetTimeZoneInformation
GetSystemTime
GetLocalTime
DuplicateHandle
GetCommandLineA
GetVersion
SetStdHandle
GetFileType
SetHandleCount
GetStdHandle
GetStartupInfoA
CreatePipe
GetExitCodeProcess
WaitForSingleObject
HeapReAlloc
HeapAlloc
```

Select Command Prompt

```
GetLastError
ReadFile
WriteFile
Sleep
GetShortPathNameA
GetModuleFileNameA
CopyFileA
ExpandEnvironmentStringsA
DeleteFileA
KERNEL32.dll
RegQueryValueExA
RegOpenKeyExA
RegSetValueExA
RegCreateKeyExA
RegDeleteValueA
CreateServiceA
CloseServiceHandle
ChangeServiceConfigA
OpenServiceA
OpenSCManagerA
DeleteService
ADVAPI32.dll
ShellExecuteA
SHELL32.dll
WS2_32.dll
ExitProcess
TerminateProcess
GetCurrentProcess
GetTimeZoneInformation
GetSystemTime
GetLocalTime
DuplicateHandle
GetCommandLineA
GetVersion
SetStdHandle
GetFileType
SetHandleCount
GetStdHandle
GetStartupInfoA
CreatePipe
GetExitCodeProcess
WaitForSingleObject
HeapReAlloc
HeapAlloc
```

Select Command Prompt

```
R6027
- not enough space for lowio initialization
R6026
- not enough space for stdio initialization
R6025
- pure virtual function call
R6024
- not enough space for _onexit/atexit table
R6019
- unable to open console device
R6018
- unexpected heap error
R6017
- unexpected multithread lock error
R6016
- not enough space for thread data
abnormal program termination
R6009
- not enough space for environment
R6008
- not enough space for arguments
R6002
- floating point not loaded
Microsoft Visual C++ Runtime Library
Runtime Error!
Program:
<program name unknown>
SunMonTueWedThuFriSat
JanFebMarAprMayJunJulAugSepOctNovDec
.com
.bat
.cmd
GetLastActivePopup
GetActiveWindow
MessageBoxA
user32.dll
PATH
CloseHandle
SetFileTime
GetFileTime
CreateFileA
GetSystemDirectoryA
GetLastError
ReadFile
```

```
Select Command Prompt
GetEnvironmentVariableA
GetVersionExA
HeapDestroy
HeapCreate
VirtualFree
HeapFree
RtlUnwind
MultiByteToWideChar
GetStringTypeA
GetStringTypeW
SetFilePointer
VirtualAlloc
LCMapStringA
LCMapStringW
GetProcAddress
LoadLibraryA
FlushFileBuffers
GetFileAttributesA
CreateProcessA
CompareStringA
CompareStringW
SetEnvironmentVariableA
Configuration
SOFTWARE\Microsoft \XPS
\kernel32.dll
 HTTP/1.0
GET
'`'`'
`'`'`
NOTHING
DOWNLOAD
UPLOAD
SLEEP
cmd.exe
 >> NUL
/c del
http://www.practicalmalwareanalysis.com
 Manager Service
.exe
%SYSTEMROOT%\system32\
k:%s h:%s p:%s per:%s
         (((((                  H
```

Which strings are possible command-line options that could guide the malware down different code paths?

The string "k:%s h:%s p:%s per:%s" appears to be a format string for parsing command-line arguments, likely guiding the malware's configuration:

- "k:" – Possibly a key or encryption parameter.
- "h:" – Likely the host (e.g., C2 server address).
- "p:" – Probably the port for network connections.
- "per:" – Could be the period or interval for beaconing/check-ins.

String indicating that something will be getting deleted

The string "/c del" (part of "cmd.exe /c del") indicates file deletion, as "del" is the Windows command for deleting files or directories.

String that suppresses output, possibly about the deletion of files

The string " > NUL" (part of "cmd.exe /c del > NUL") redirects command output to the null device (NUL), suppressing any console messages or errors, which could hide traces of deletions or other actions.

Strings that look like possible menu items for adversary interaction

If this is C2 malware, the following strings resemble commands or menu options an attacker might select/send for remote control:

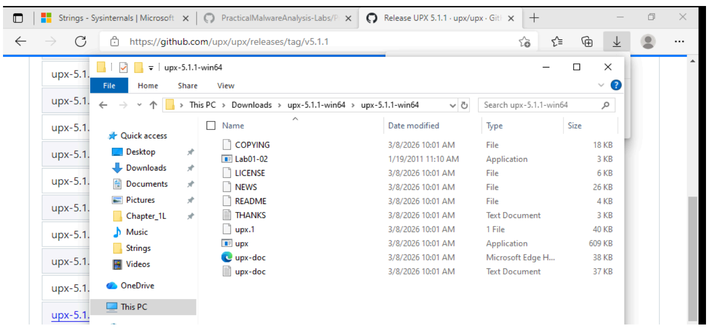
- "NOTHING" – Likely a no-op or idle command.
- "UPLOAD" – Probably for uploading files or data from the victim machine.
- "SLEEP" – Could instruct the malware to delay actions or enter a dormant state.

Hypothesis

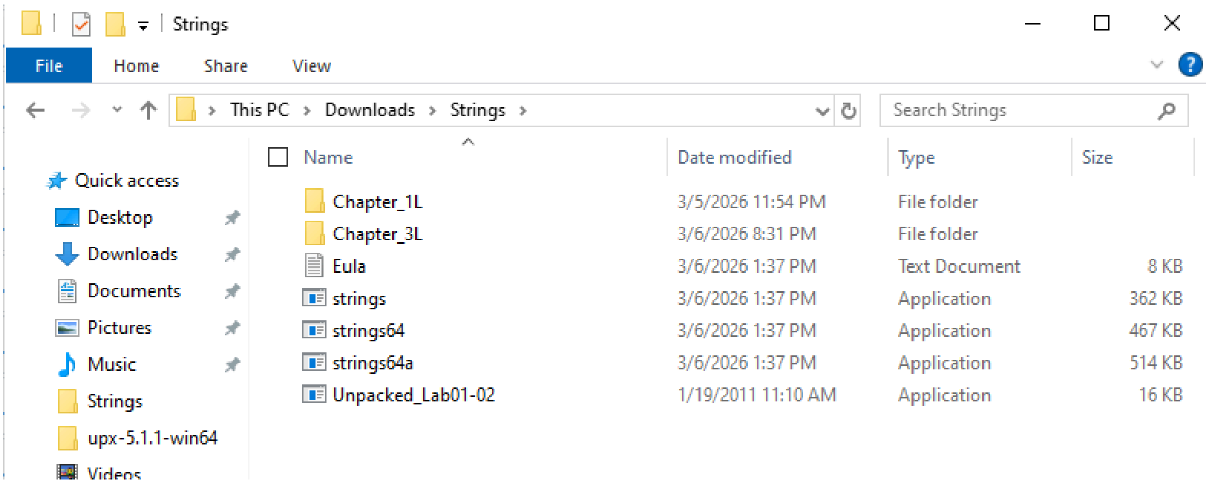
This seems to be a persistent backdoor trojan. It installs itself as "Hider Service" (masquerading as a legitimate service) in %SYSTEMROOT%\system32\ for evasion, using APIs like CreateServiceA, OpenSCManagerA, and registry modifications (e.g., under SOFTWARE\Microsoft\XPS) for auto-start. It supports configurable parameters via command-line (e.g., host, port, key, period) and establishes HTTP/1.0 connections to http://www.practicalmalwareanalysis.com for C2, using GET requests. Commands like UPLOAD (file exfiltration), SLEEP (timing control), and NOTHING (placeholder) enable attacker interaction, while shell execution via cmd.exe /c (with output suppression via > NUL and deletion via del) allows arbitrary commands and cleanup. File operations (CopyFileA, ReadFile), process creation (CreateProcessA), and environment handling (GetEnvironmentVariableA) support payload deployment or data theft. Runtime errors suggest it's compiled with Visual C++ and may crash under certain conditions, but overall, it's a basic RAT focused on stealth and remote access.

Lab Exercise 15.02: UPX

1a-1c

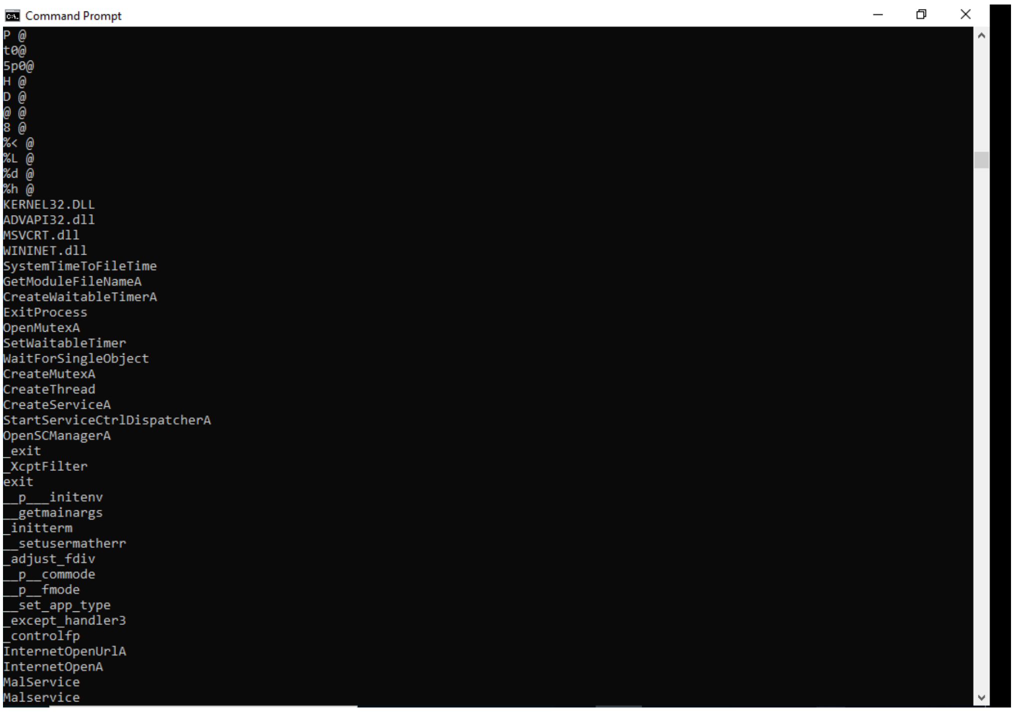


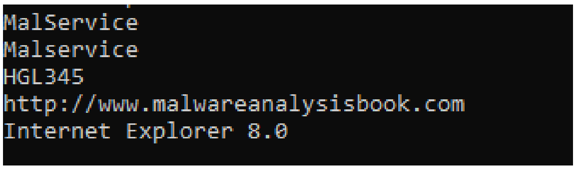
```
Select C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19044.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Users\potato\Downloads\upx-5.1.1-win64\upx-5.1.1-win64>upx -o Unpacked_Lab01-02.exe -d Lab01-02.exe
                       Ultimate Packer for eXecutables
                          Copyright (C) 1996 - 2026
UPX 5.1.1       Markus Oberhumer, Laszlo Molnar & John Reiser    Mar 5th 2026

        File size         Ratio      Format      Name
   --------------------   ------   -----------   -----------
     16384 <-      3072   18.75%    win32/pe     Unpacked_Lab01-02.exe

Unpacked 1 file.
```



2a-2d

```
C:\Users\potato\Downloads\Strings>strings64 Unpacked_Lab01-02.exe | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
@jj
h(0@
  @
Vh(0@
, @
@jjjj
L$,j
@jjj
@jjj
T$
$ @
( @
u+W
=0 @
jjj
@jj
VWj
jjj
hT0@
t @
=p @
h00@
 SVW
` @
\ @
|0@
X @
x0@
T @
=10@
P @
```
Microsoft Store





The URL "http://www.malwareanalysisbook.com" is now visible

Some related functions now visible:

Functions related to the URL and network communication include:

- InternetOpenUrlA: Opens a URL for HTTP access.
- InternetOpenA: Initializes an internet session (e.g., for HTTP requests).

Other newly visible functions support overall behavior:

- CreateServiceA: Creates a Windows service.
- OpenSCManagerA: Opens the Service Control Manager for service operations.
- StartServiceCtrlDispatcherA: Dispatches service control requests.
- CreateMutexA and OpenMutexA: Manage mutexes for single-instance execution.
- CreateThread: Spawns threads (e.g., for background tasks).
- SetWaitableTimer and WaitForSingleObject: Handle timing/delays, possibly for beaconing intervals.

These indicate service persistence and timed network operations.

Visible DLLs:

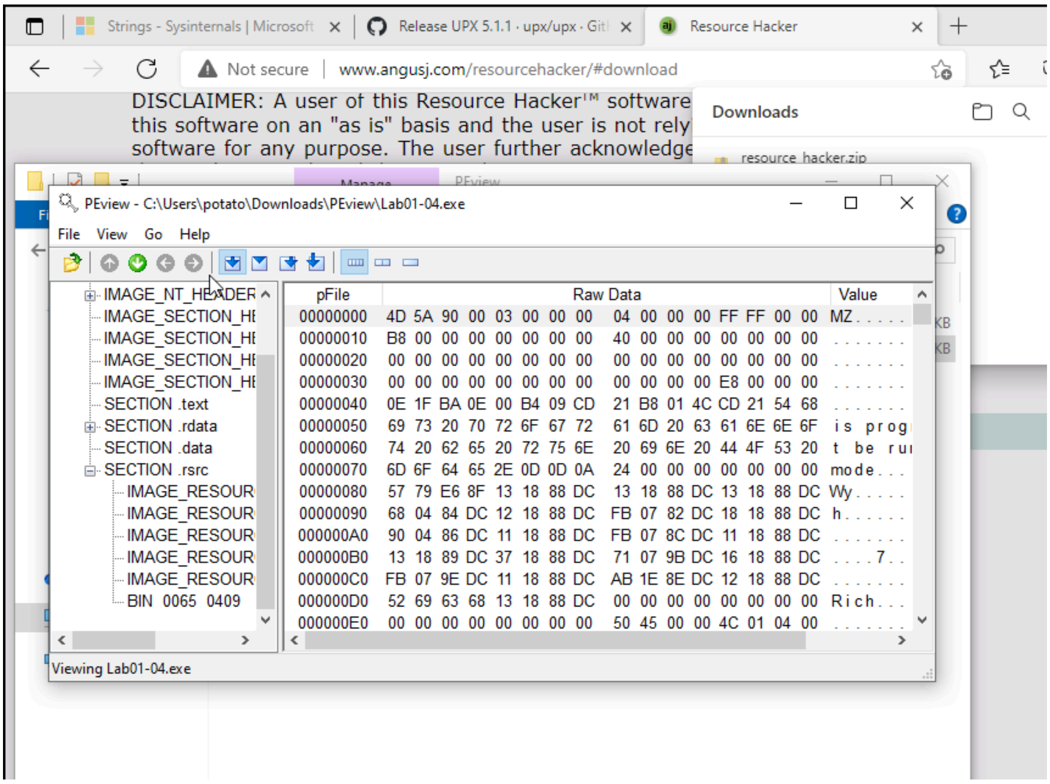
- Network functions (InternetOpenUrlA, InternetOpenA) come from WININET.DLL.
- Service functions (CreateServiceA, OpenSCManagerA, StartServiceCtrlDispatcherA) come from ADVAPI32.DLL.
- Mutex and threading functions (CreateMutexA, OpenMutexA, CreateThread, SetWaitableTimer, WaitForSingleObject) come from KERNEL32.DLL.

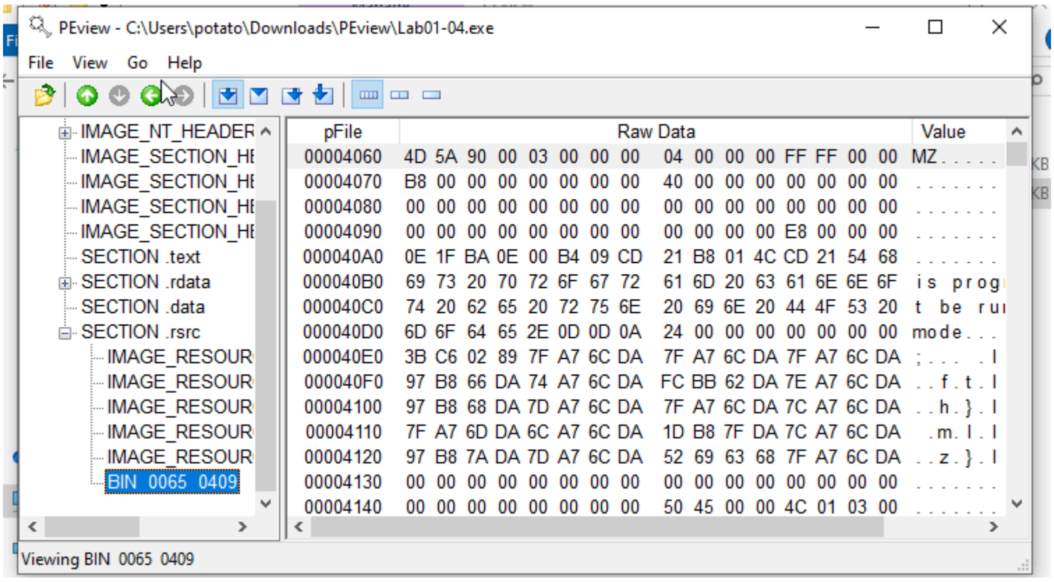
Hypothesis

The unpacked Lab01-02.exe is a basic trojan installer/downloader. It creates and starts a Windows service named "MalService" (using ADVAPI32.DLL APIs) for persistence, ensures single-instance via a mutex (possibly "HGL345"), and spawns threads with timers for delayed execution. It then initializes an internet session mimicking "Internet Explorer 8.0" as the user agent and connects to http://www.malwareanalysisbook.com, likely to download additional payloads or receive commands. The original UPX packing hid these details to evade static analysis, but unpacking exposes its straightforward backdoor-like functionality.

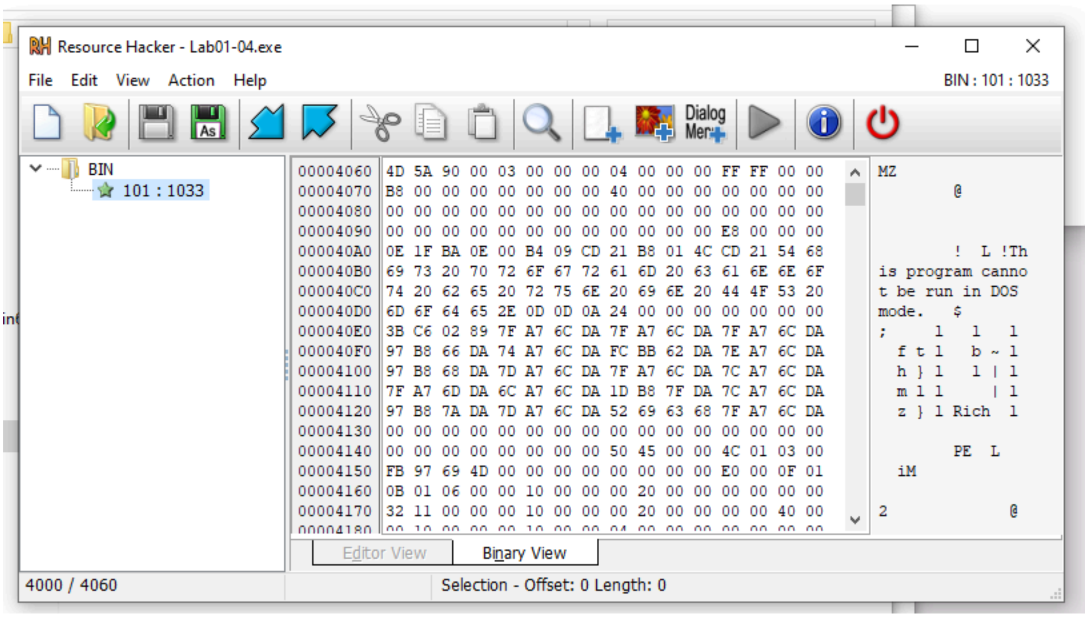
Lab Exercise 15.03: PEview and Resource Hacker

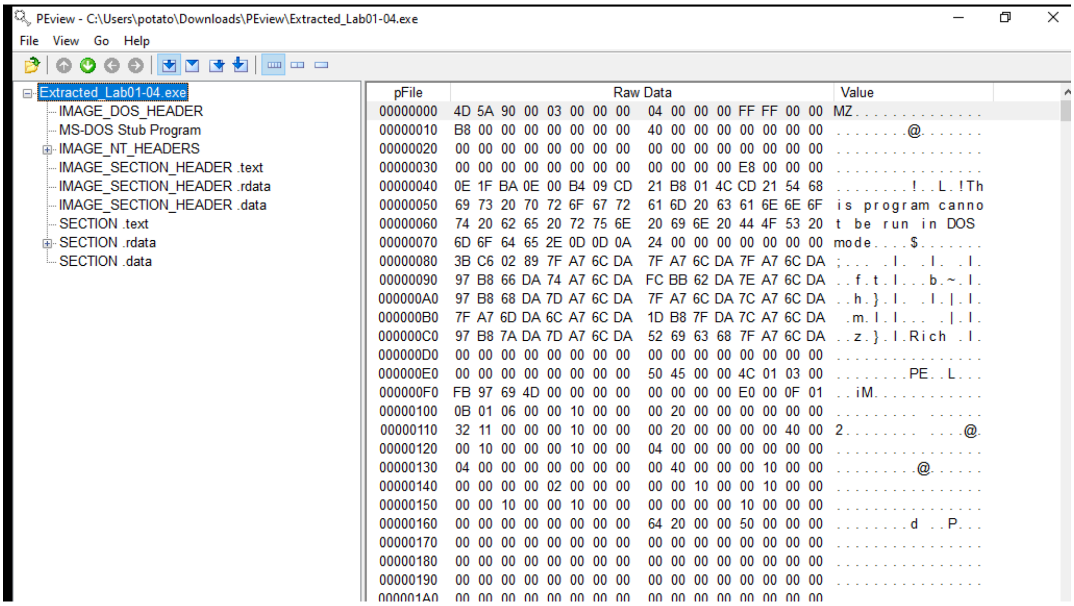
1d-1f

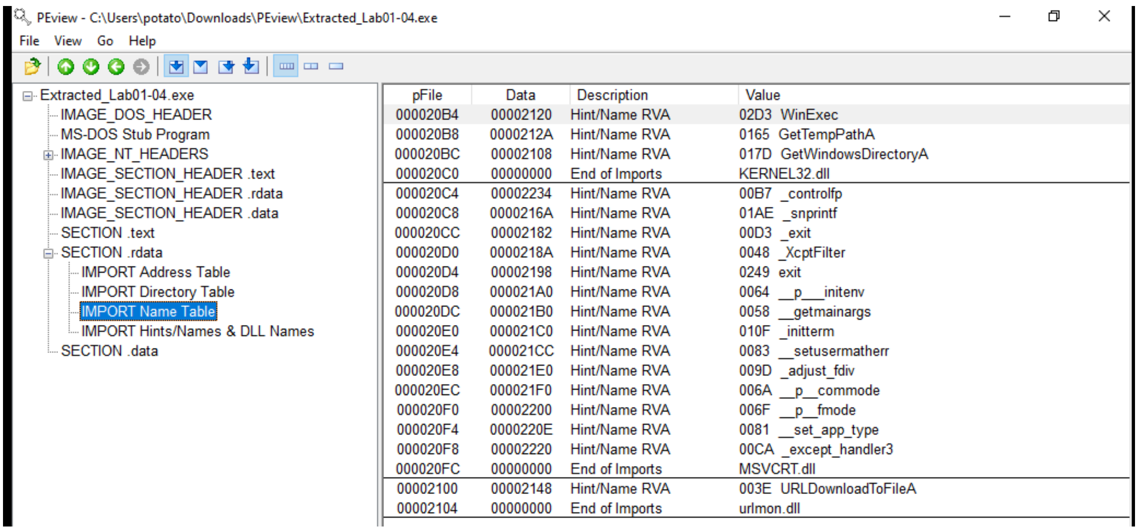

PEview - C:\Users\potato\Downloads\PEview\Lab01-04.exe
Viewing Lab01-04.exe


PEview - C:\Users\potato\Downloads\PEview\Lab01-04.exe
Viewing BIN 0065 0409

2a-2e


Resource Hacker - Lab01-04.exe
File Edit View Action Help
BIN : 101 : 1033
BIN
101 : 1033
00004060 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ
00004070 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 @
00004080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00004090 00 00 00 00 00 00 00 00 00 00 00 00 E8 00 00 00
000040A0 0E 1F BA 0E 00 B4 09 CD 21 B8 01 4C CD 21 54 68 ! L !Th
000040B0 69 73 20 70 72 6F 67 72 61 6D 20 63 61 6E 6E 6F is program canno
000040C0 74 20 62 65 20 72 75 6E 20 69 6E 20 44 4F 53 20 t be run in DOS
000040D0 6D 6F 64 65 2E 0D 0D 0A 24 00 00 00 00 00 00 00 mode. $
000040E0 3B C6 02 89 7F A7 6C DA 7F A7 6C DA 7F A7 6C DA
000040F0 97 B8 66 DA 74 A7 6C DA FC BB 62 DA 7E A7 6C DA
00004100 97 B8 68 DA 7D A7 6C DA 7F A7 6C DA 7C A7 6C DA
00004110 7F A7 6D DA 6C A7 6C DA 1D B8 7F DA 7C A7 6C DA
00004120 97 B8 7A DA 7D A7 6C DA 52 69 63 68 7F A7 6C DA Rich
00004130 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00004140 00 00 00 00 00 00 00 00 50 45 00 00 4C 01 03 00 PE L
00004150 FB 97 69 4D 00 00 00 00 00 00 00 00 E0 00 0F 01 iM
00004160 0B 01 06 00 00 10 00 00 00 20 00 00 00 00 00 00
00004170 32 11 00 00 00 10 00 00 00 20 00 00 00 00 40 00 2 @
Editor View
Binary View
4000 / 4060
Selection - Offset: 0 Length: 0


PEview - C:\Users\potato\Downloads\PEview\Extracted_Lab01-04.exe
File View Go Help
Extracted_Lab01-04.exe
IMAGE_DOS_HEADER
MS-DOS Stub Program
IMAGE_NT_HEADERS
IMAGE_SECTION_HEADER .text
IMAGE_SECTION_HEADER .rdata
IMAGE_SECTION_HEADER .data
SECTION .text
SECTION .rdata
SECTION .data
pFile Raw Data Value
00000000 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ
00000010 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 @
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000030 00 00 00 00 00 00 00 00 00 00 00 00 E8 00 00 00
00000040 0E 1F BA 0E 00 B4 09 CD 21 B8 01 4C CD 21 54 68 !..L.!Th
00000050 69 73 20 70 72 6F 67 72 61 6D 20 63 61 6E 6E 6F is program canno
00000060 74 20 62 65 20 72 75 6E 20 69 6E 20 44 4F 53 20 t be run in DOS
00000070 6D 6F 64 65 2E 0D 0D 0A 24 00 00 00 00 00 00 00 mode....$
00000080 3B C6 02 89 7F A7 6C DA 7F A7 6C DA 7F A7 6C DA
00000090 97 B8 66 DA 74 A7 6C DA FC BB 62 DA 7E A7 6C DA
000000A0 97 B8 68 DA 7D A7 6C DA 7F A7 6C DA 7C A7 6C DA
000000B0 7F A7 6D DA 6C A7 6C DA 1D B8 7F DA 7C A7 6C DA
000000C0 97 B8 7A DA 7D A7 6C DA 52 69 63 68 7F A7 6C DA Rich
000000D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
000000E0 00 00 00 00 00 00 00 00 50 45 00 00 4C 01 03 00 PE..L
000000F0 FB 97 69 4D 00 00 00 00 00 00 00 00 E0 00 0F 01 iM
00000100 0B 01 06 00 00 10 00 00 00 20 00 00 00 00 00 00
00000110 32 11 00 00 00 10 00 00 00 20 00 00 00 00 40 00 2 @
00000120 00 10 00 00 00 10 00 00 04 00 00 00 00 00 00 00
00000130 04 00 00 00 00 00 00 00 00 40 00 00 00 10 00 00 @
00000140 00 00 00 00 02 00 00 00 00 00 10 00 00 10 00 00
00000150 00 00 10 00 00 10 00 00 00 00 00 00 10 00 00 00
00000160 00 00 00 00 00 00 00 00 64 20 00 00 50 00 00 00 d P
00000170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00


PEview - C:\Users\potato\Downloads\PEview\Extracted_Lab01-04.exe
File View Go Help
Extracted_Lab01-04.exe
IMAGE_DOS_HEADER
MS-DOS Stub Program
IMAGE_NT_HEADERS
IMAGE_SECTION_HEADER .text
IMAGE_SECTION_HEADER .rdata
IMAGE_SECTION_HEADER .data
SECTION .text
SECTION .rdata
IMPORT Address Table
IMPORT Directory Table
IMPORT Name Table
IMPORT Hints/Names & DLL Names
SECTION .data
pFile Data Description Value
000020B4 00002120 Hint/Name RVA 02D3 WinExec
000020B8 0000212A Hint/Name RVA 0165 GetTempPathA
000020BC 00002108 Hint/Name RVA 017D GetWindowsDirectoryA
000020C0 00000000 End of Imports KERNEL32.dll
000020C4 00002234 Hint/Name RVA 00B7 _controlfp
000020C8 0000216A Hint/Name RVA 01AE _snprintf
000020CC 00002182 Hint/Name RVA 00D3 _exit
000020D0 0000218A Hint/Name RVA 0048 _XcptFilter
000020D4 00002198 Hint/Name RVA 0249 exit
000020D8 000021A0 Hint/Name RVA 0064 __p___initenv
000020DC 000021B0 Hint/Name RVA 0058 __getmainargs
000020E0 000021C0 Hint/Name RVA 010F _initterm
000020E4 000021CC Hint/Name RVA 0083 __setusermatherr
000020E8 000021E0 Hint/Name RVA 009D _adjust_fdiv
000020EC 000021F0 Hint/Name RVA 006A __p__commode
000020F0 00002200 Hint/Name RVA 006F __p__fmode
000020F4 0000220E Hint/Name RVA 0081 __set_app_type
000020F8 00002220 Hint/Name RVA 00CA _except_handler3
000020FC 00000000 End of Imports MSVCRT.dll
00002100 00002148 Hint/Name RVA 003E URLDownloadToFileA
00002104 00000000 End of Imports urlmon.dll

4a-4c

C:\Windows\System32\cmd.exe

```
C:\Users\potato\Downloads\PEview>strings64 -n 4 Extracted_Lab01-04.exe | more

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Rich
.text
`.rdata
@.data
h$0@
Rh<0@
QhD0@
%L @
hX @
 SVW
=x0@
%, @
%D @
GetWindowsDirectoryA
WinExec
GetTempPathA
KERNEL32.dll
URLDownloadToFileA
urlmon.dll
_snprintf
MSVCRT.dll
_exit
_XcptFilter
exit
__p___initenv
__getmainargs
_initterm
__setusermatherr
_adjust_fdiv
__p__commode
__p__fmode
__set_app_type
_except_handler3
_controlfp
\winup.exe
%s%s
\system32\wupdmgrd.exe
%s%s
http://www.practicalmalwareanalysis.com/updater.exe
```

Related URL strings found:

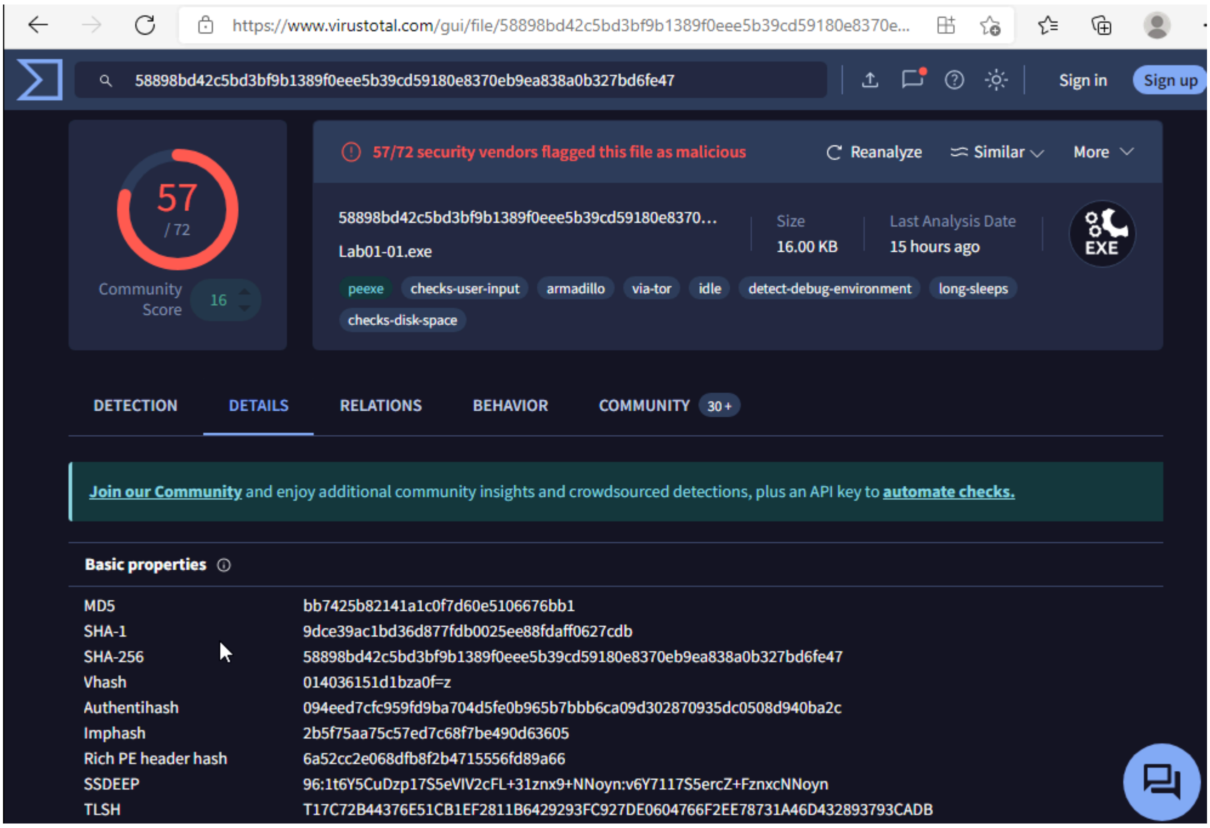
- http://www.practicalmalwareanalysis.com/updater.exe
- Other URL-adjacent or path-related strings include: %system32\wupdmgrd.exe (and similar paths like \system32\wupdmgrd.exe, \winup.exe)

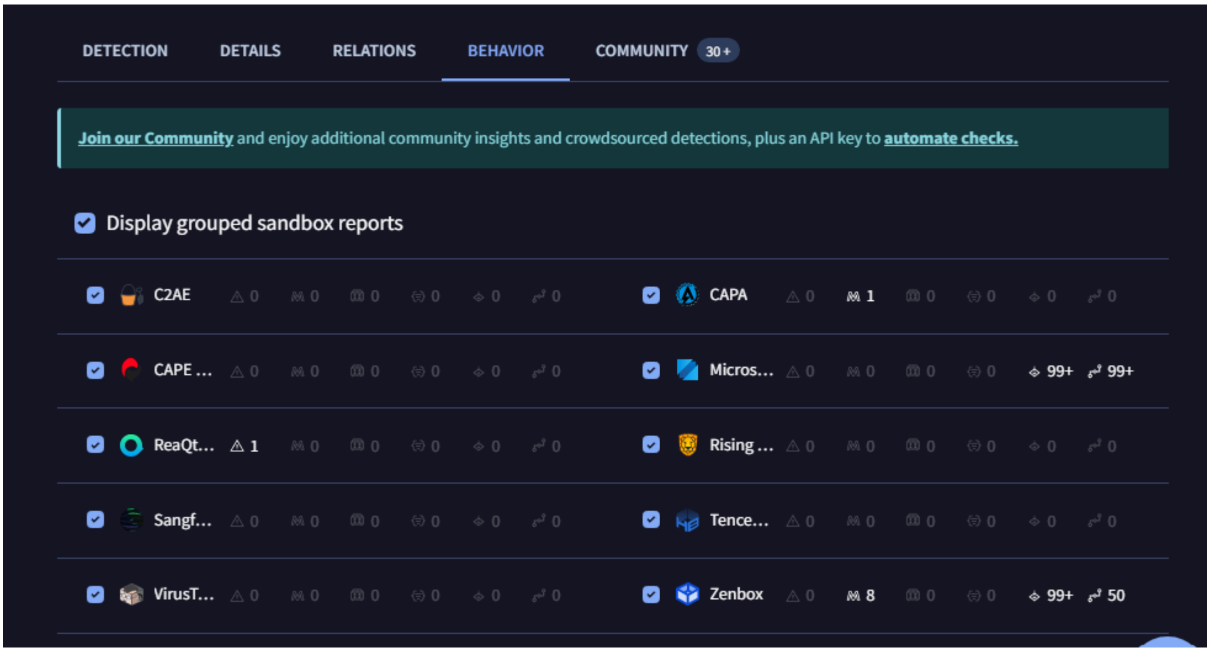
Strings related to updates:

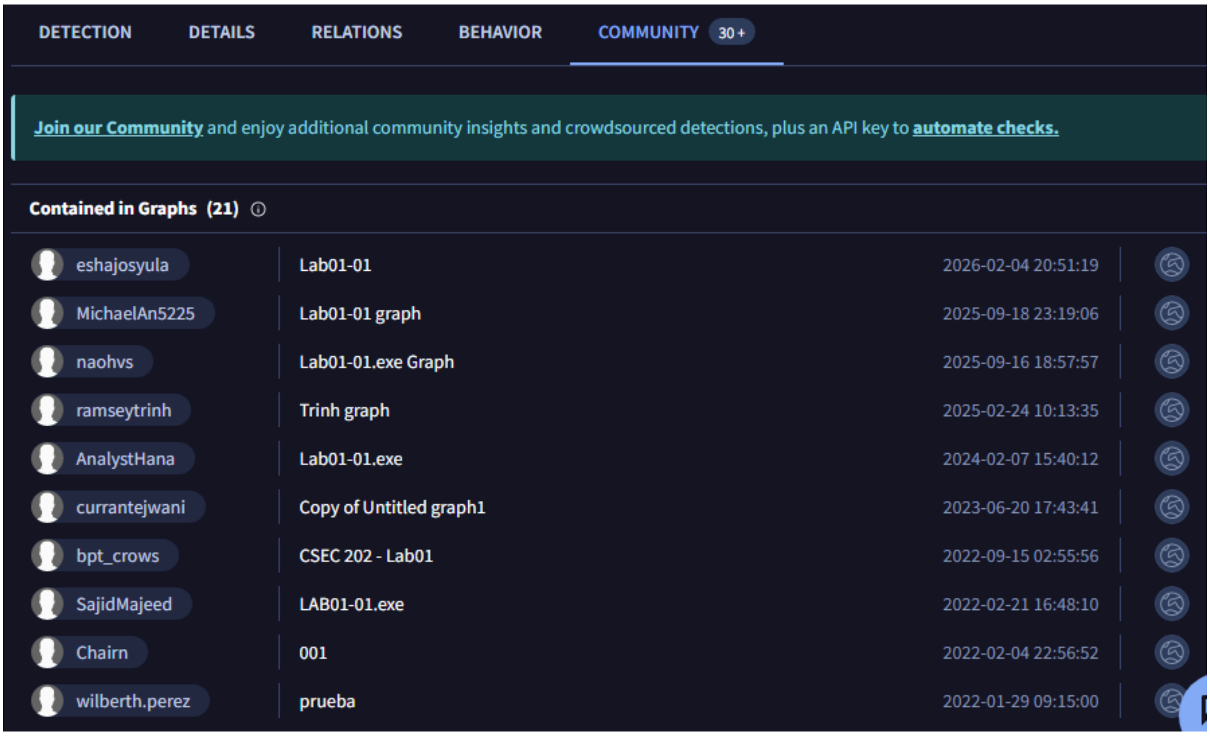
- updater.exe (in the full URL above) — strongly suggests this binary downloads and possibly executes something positioned as an "updater."
- wupdmgrd.exe (appears multiple times, e.g., \system32\wupdmgrd.exe) — a slight misspelling/variation of the legitimate wupdmgr.exe (Windows Update Manager). This is a classic trick to blend in with system update processes.
- winup.exe (or \winup.exe) — another fake/forged name mimicking Windows Update-related files.

Lab Exercise 15.04: VirusTotal

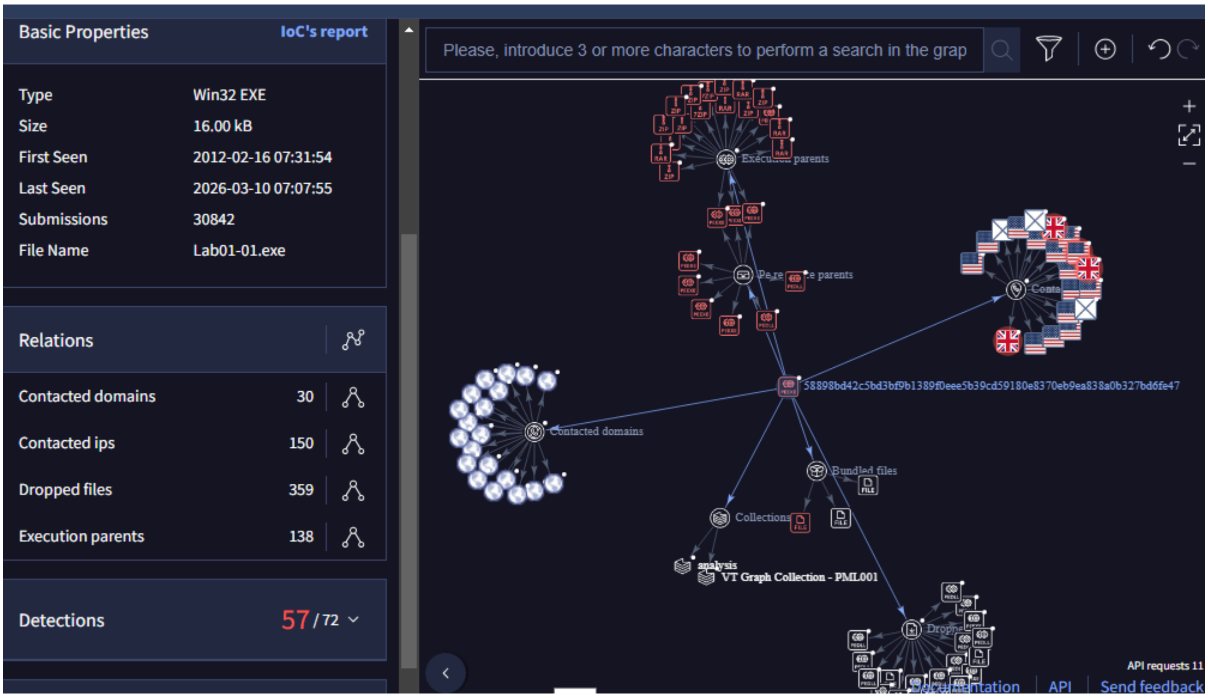
1a-1e

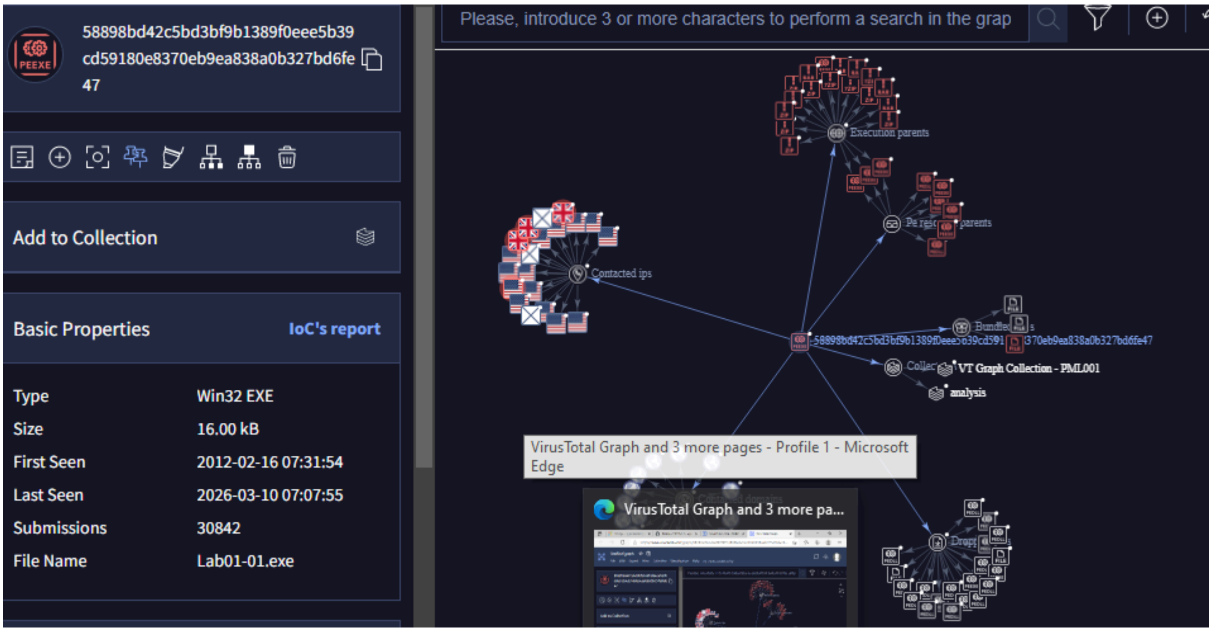




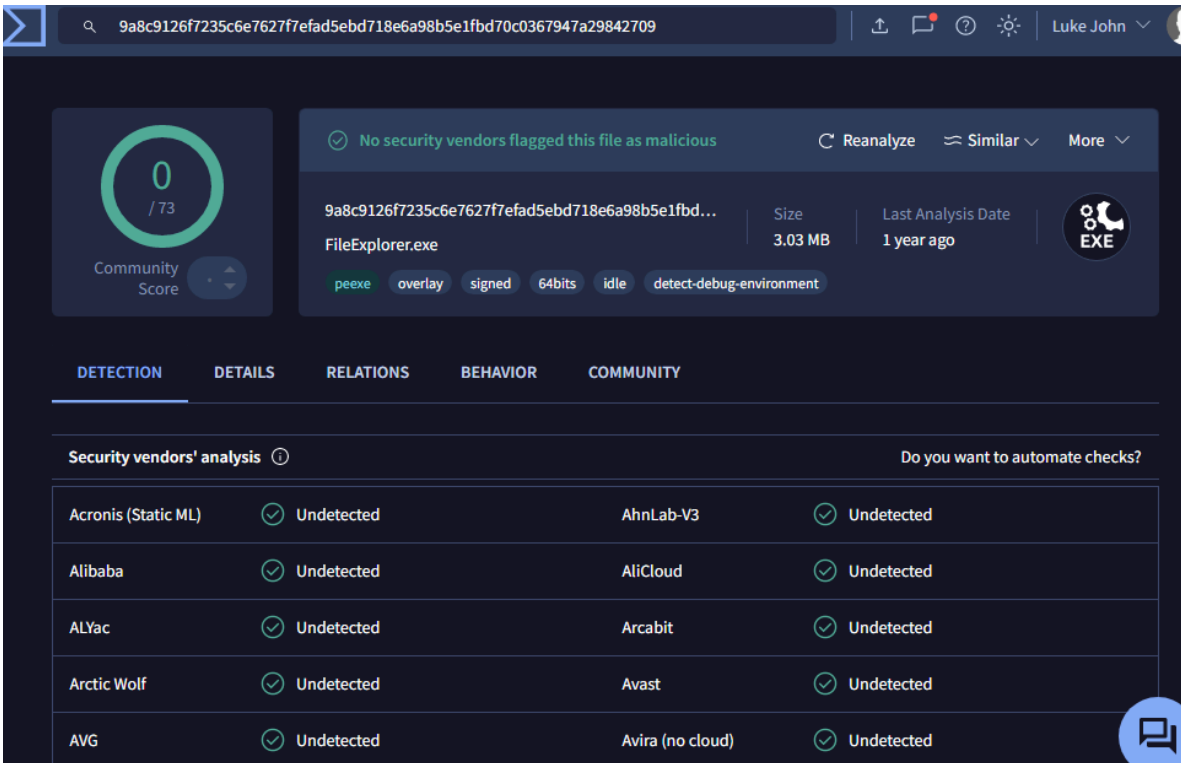

DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 30+
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Contained in Graphs (21)
eshajosyula Lab01-01 2026-02-04 20:51:19
MichaelAn5225 Lab01-01 graph 2025-09-18 23:19:06
naohvs Lab01-01.exe Graph 2025-09-16 18:57:57
ramseytrinh Trinh graph 2025-02-24 10:13:35
AnalystHana Lab01-01.exe 2024-02-07 15:40:12
currantejwani Copy of Untitled graph1 2023-06-20 17:43:41
bpt_crows CSEC 202 - Lab01 2022-09-15 02:55:56
SajidMajeed LAB01-01.exe 2022-02-21 16:48:10
Chairn 001 2022-02-04 22:56:52
wilberth.perez prueba 2022-01-29 09:15:00

2a-2c


Basic Properties
IoC's report
Type Win32 EXE
Size 16.00 kB
First Seen 2012-02-16 07:31:54
Last Seen 2026-03-10 07:07:55
Submissions 30842
File Name Lab01-01.exe
Relations
Contacted domains 30
Contacted ips 150
Dropped files 359
Execution parents 138
Detections 57 / 72
Please, introduce 3 or more characters to perform a search in the grap
Contacted domains
Collections
Bundled files
VT Graph Collection - PML001
58898bd42c5bd3bf9b1389f0eee5b39cd59180e8370eb9ea838a0b327bd6fe47
API
Send feedback

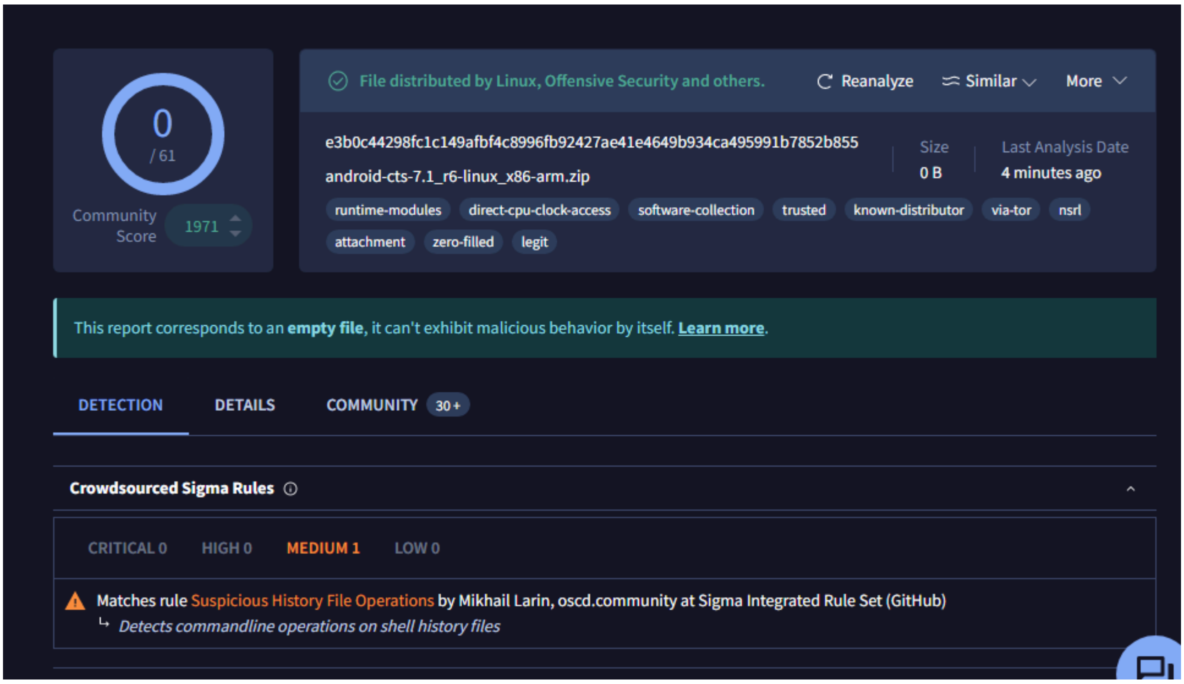


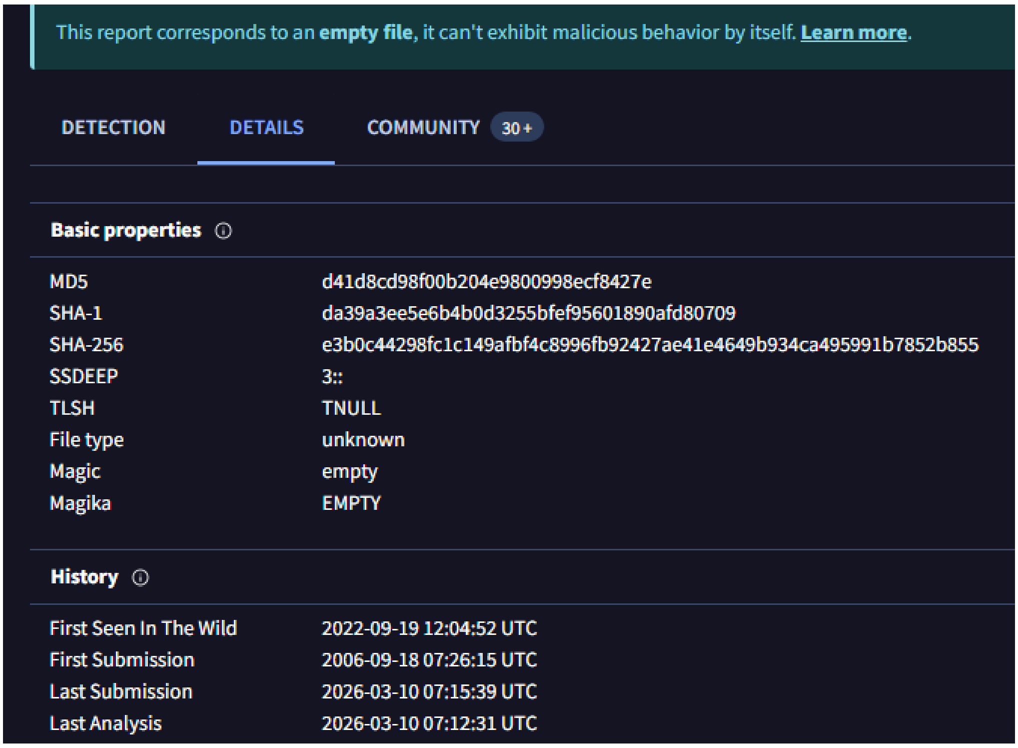
3a-3b (Uploaded harmless FileExplorer.exe instead of malware)



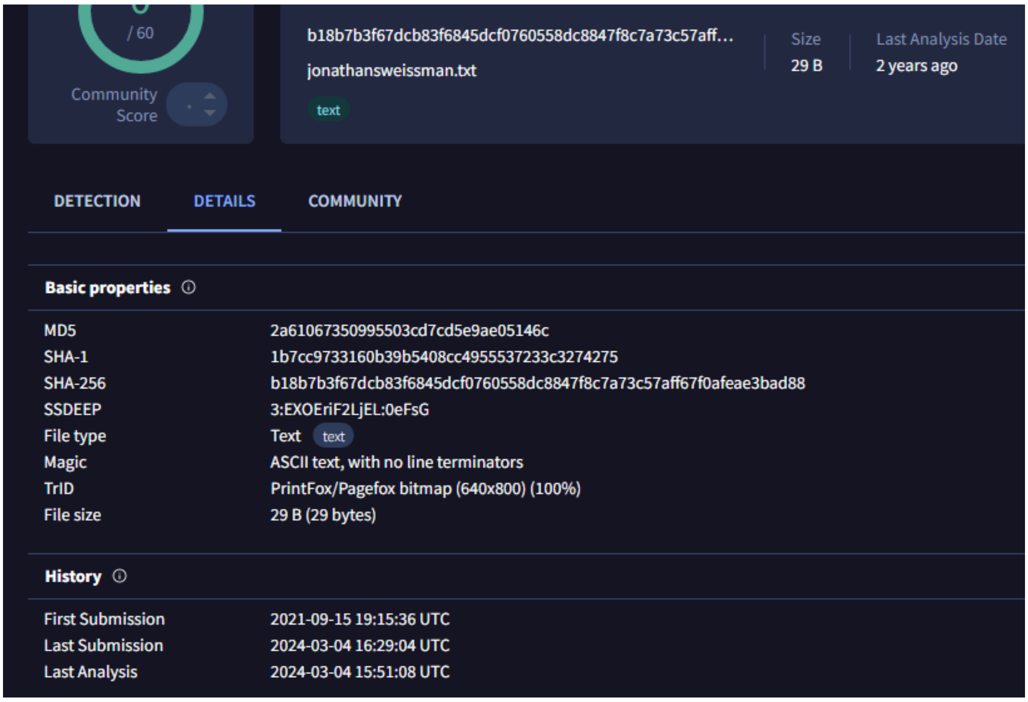
No vendors flag this as suspicious or malicious

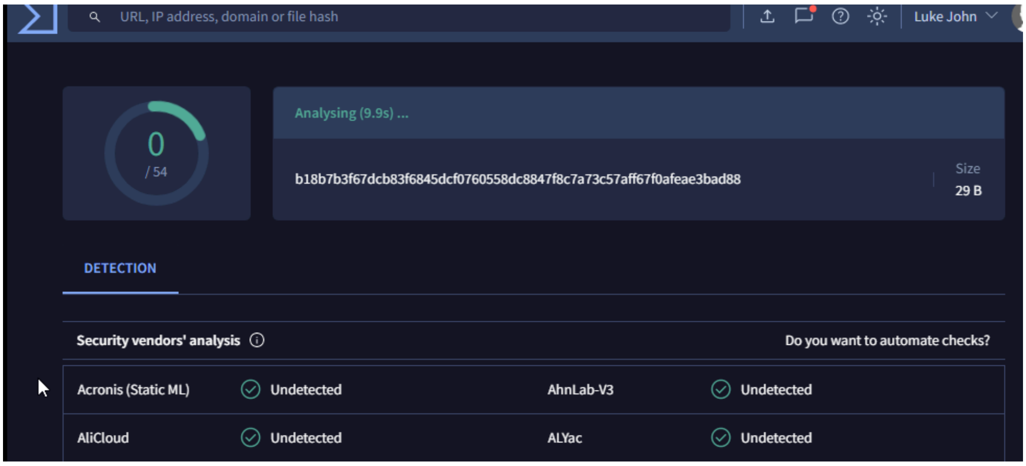
4a-4b

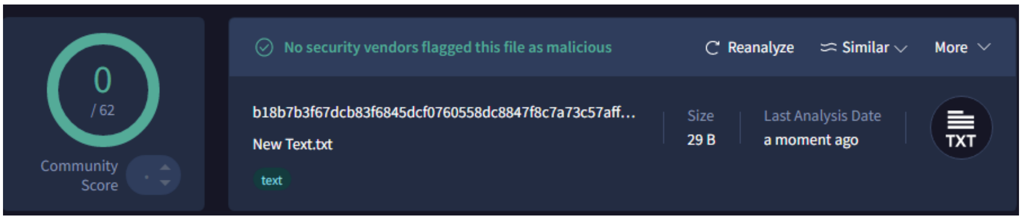

0
/ 61
Community Score
1971
File distributed by Linux, Offensive Security and others.
Reanalyze
Similar
More
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
android-cts-7.1_r6-linux_x86-arm.zip
Size
0 B
Last Analysis Date
4 minutes ago
runtime-modules direct-cpu-clock-access software-collection trusted known-distributor via-tor nsrl
attachment zero-filled legit
This report corresponds to an empty file, it can't exhibit malicious behavior by itself. Learn more.
DETECTION
DETAILS
COMMUNITY 30+
Crowdsourced Sigma Rules
CRITICAL 0 HIGH 0 MEDIUM 1 LOW 0
Matches rule Suspicious History File Operations by Mikhail Larin, oscd.community at Sigma Integrated Rule Set (GitHub)
Detects commandline operations on shell history files


This report corresponds to an empty file, it can't exhibit malicious behavior by itself. Learn more.
DETECTION
DETAILS
COMMUNITY 30+
Basic properties
MD5 d41d8cd98f00b204e9800998ecf8427e
SHA-1 da39a3ee5e6b4b0d3255bfef95601890afd80709
SHA-256 e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
SSDEEP 3::
TLSH TNULL
File type unknown
Magic empty
Magika EMPTY
History
First Seen In The Wild 2022-09-19 12:04:52 UTC
First Submission 2006-09-18 07:26:15 UTC
Last Submission 2026-03-10 07:15:39 UTC
Last Analysis 2026-03-10 07:12:31 UTC

5a-5d

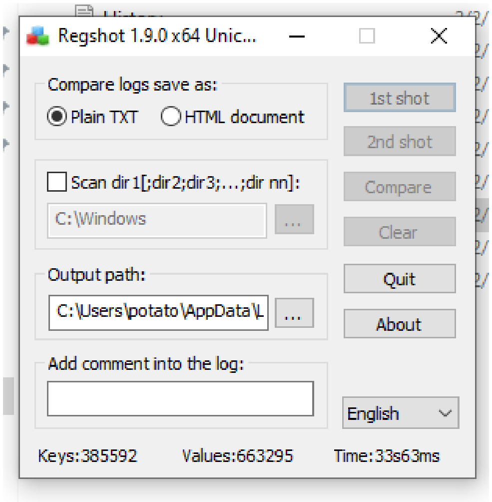

/ 60
Community Score
b18b7b3f67dcb83f6845dcf0760558dc8847f8c7a73c57aff...
jonathansweissman.txt
text
Size
29 B
Last Analysis Date
2 years ago
DETECTION
DETAILS
COMMUNITY
Basic properties
MD5
2a61067350995503cd7cd5e9ae05146c
SHA-1
1b7cc9733160b39b5408cc4955537233c3274275
SHA-256
b18b7b3f67dcb83f6845dcf0760558dc8847f8c7a73c57aff67f0afeae3bad88
SSDEEP
3:EXOEriF2LjEL:0eFsG
File type
Text
text
Magic
ASCII text, with no line terminators
TrID
PrintFox/Pagefox bitmap (640x800) (100%)
File size
29 B (29 bytes)
History
First Submission
2021-09-15 19:15:36 UTC
Last Submission
2024-03-04 16:29:04 UTC
Last Analysis
2024-03-04 15:51:08 UTC

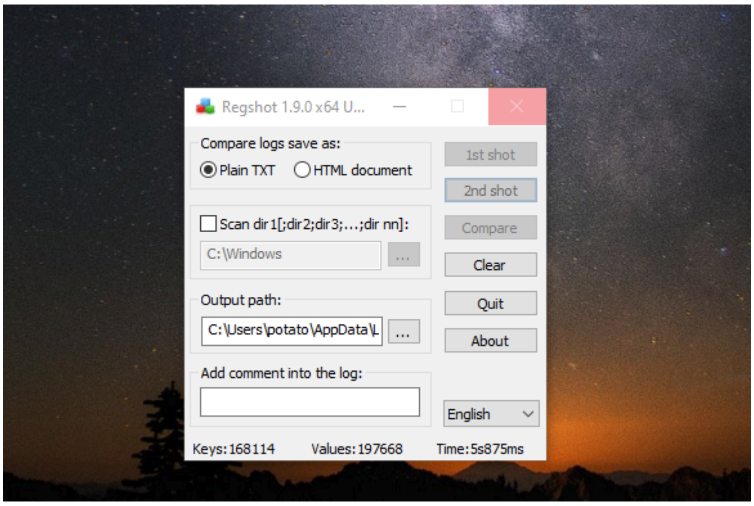

URL, IP address, domain or file hash
Luke John
0
/ 54
Analysing (9.9s) ...
b18b7b3f67dcb83f6845dcf0760558dc8847f8c7a73c57aff67f0afeae3bad88
Size
29 B
DETECTION
Security vendors' analysis
Do you want to automate checks?
Acronis (Static ML)
Undetected
AhnLab-V3
Undetected
AliCloud
Undetected
ALYac
Undetected

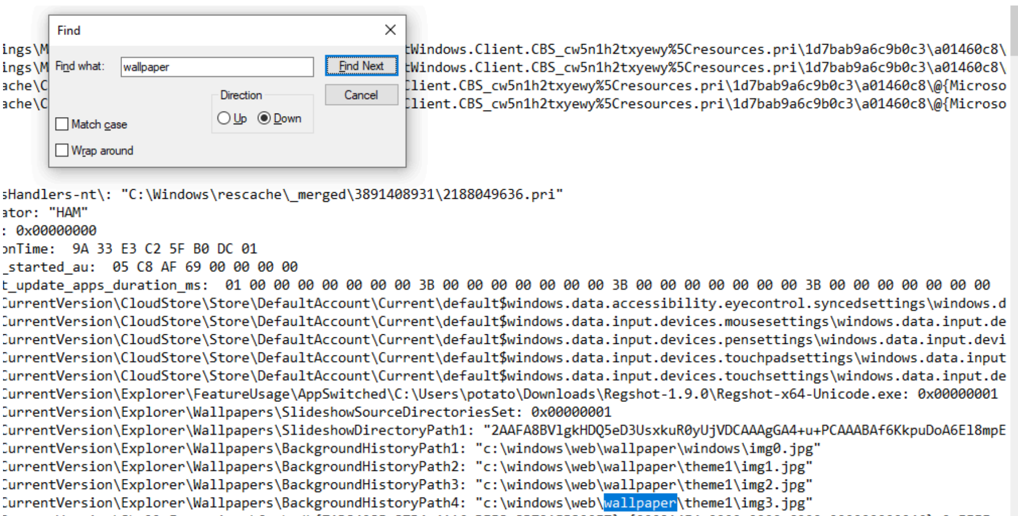

0
/ 62
Community Score
No security vendors flagged this file as malicious
Reanalyze
Similar
More
b18b7b3f67dcb83f6845dcf0760558dc8847f8c7a73c57aff...
New Text.txt
text
Size
29 B
Last Analysis Date
a moment ago
TXT

Lab Exercise 15.05: Regshot

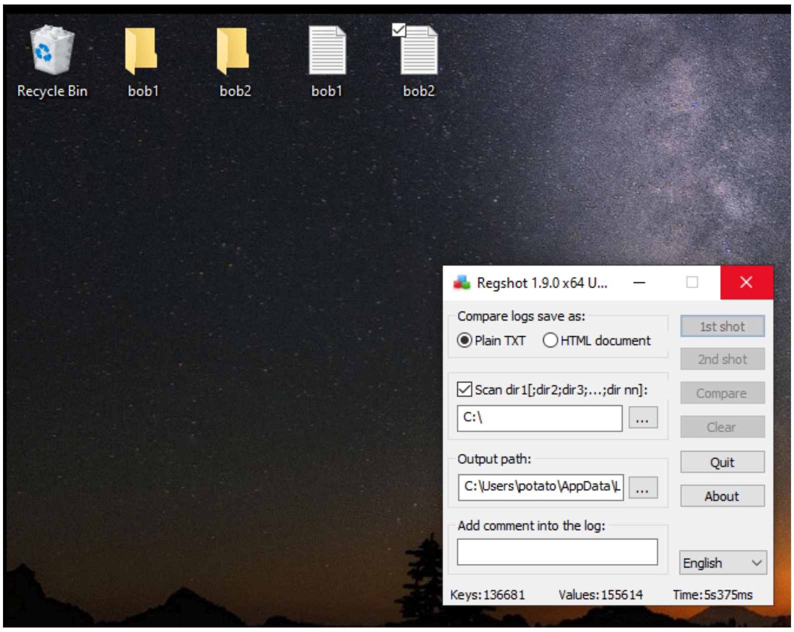
1d-1f

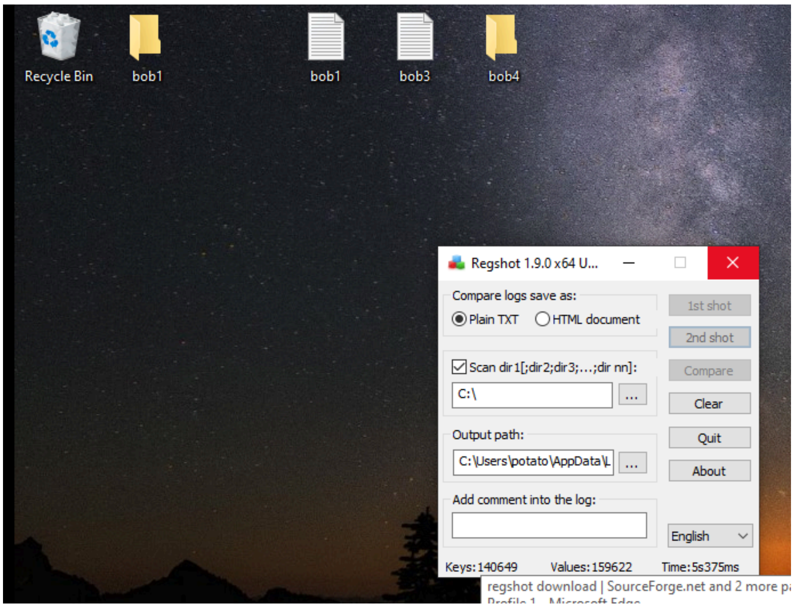


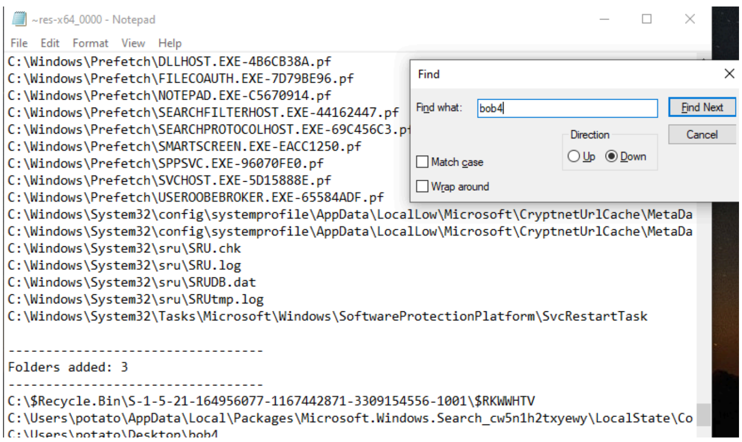




2a-2m


Recycle Bin
bob1
bob2
bob1
bob2
Regshot 1.9.0 x64 U...
Compare logs save as:
Plain TXT
HTML document
1st shot
2nd shot
Scan dir1[;dir2;dir3;...;dir nn]:
Compare
C:\
Clear
Output path:
Quit
C:\Users\potato\AppData\L
About
Add comment into the log:
English
Keys:136681
Values:155614
Time:5s375ms

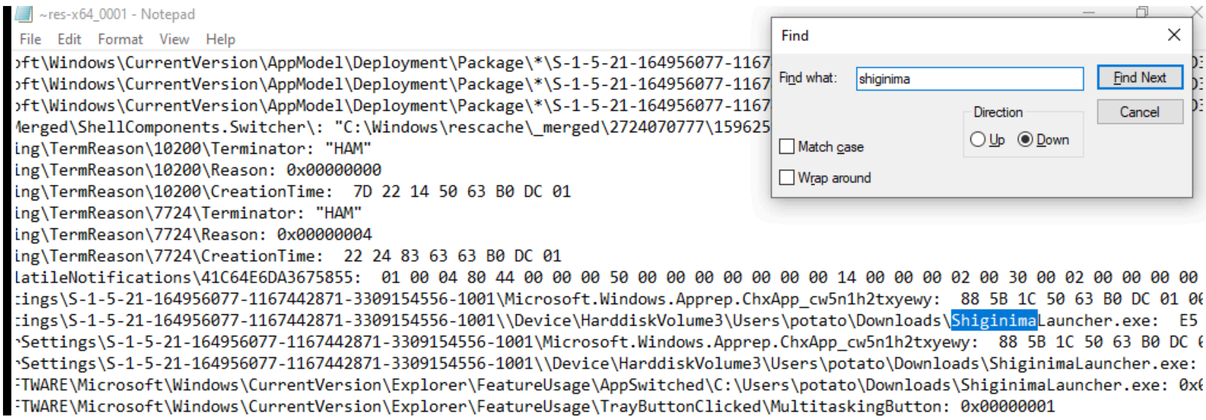

Recycle Bin
bob1
bob1
bob3
bob4
Regshot 1.9.0 x64 U...
Compare logs save as:
Plain TXT
HTML document
1st shot
2nd shot
Scan dir1[;dir2;dir3;...;dir nn]:
Compare
C:\
Clear
Output path:
Quit
C:\Users\potato\AppData\L
About
Add comment into the log:
English
Keys:140649
Values:159622
Time:5s375ms
regshot download | SourceForge.net and 2 more p


~res-x64_0000 - Notepad
File Edit Format View Help
C:\Windows\Prefetch\DLLHOST.EXE-4B6CB38A.pf
C:\Windows\Prefetch\FILECOAUTH.EXE-7D79BE96.pf
C:\Windows\Prefetch\NOTEPAD.EXE-C5670914.pf
C:\Windows\Prefetch\SEARCHFILTERHOST.EXE-44162447.pf
C:\Windows\Prefetch\SEARCHPROTOCOLHOST.EXE-69C456C3.pf
C:\Windows\Prefetch\SMARTSCREEN.EXE-EACC1250.pf
C:\Windows\Prefetch\SPPSVC.EXE-96070FE0.pf
C:\Windows\Prefetch\SVCHOST.EXE-5D15888E.pf
C:\Windows\Prefetch\USEROOBEBROKER.EXE-65584ADF.pf
C:\Windows\System32\config\systemprofile\AppData\LocalLow\Microsoft\CryptnetUrlCache\MetaDa
C:\Windows\System32\config\systemprofile\AppData\LocalLow\Microsoft\CryptnetUrlCache\MetaDa
C:\Windows\System32\sru\SRU.chk
C:\Windows\System32\sru\SRU.log
C:\Windows\System32\sru\SRUDB.dat
C:\Windows\System32\sru\SRUtmp.log
C:\Windows\System32\Tasks\Microsoft\Windows\SoftwareProtectionPlatform\SvcRestartTask
----
Folders added: 3

C:\$Recycle.Bin\S-1-5-21-164956077-1167442871-3309154556-1001\$RKWWHTV
C:\Users\potato\AppData\Local\Packages\Microsoft.Windows.Search_cw5n1h2txyewy\LocalState\Co
Find
Find what:
bob4
Find Next
Cancel
Direction
Up
Down
Match case
Wrap around

3a-3h

Download




```
ɔft\Windows\CurrentVersion\AppModel\Deployment\Package\*\S-1-5-21-164956077-1167
ɔft\Windows\CurrentVersion\AppModel\Deployment\Package\*\S-1-5-21-164956077-1167
ɔft\Windows\CurrentVersion\AppModel\Deployment\Package\*\S-1-5-21-164956077-1167
Merged\ShellComponents.Switcher\: "C:\Windows\rescache\_merged\2724070777\159625
ing\TermReason\10200\Terminator: "HAM"
ing\TermReason\10200\Reason: 0x00000000
ing\TermReason\10200\CreationTime:  7D 22 14 50 63 B0 DC 01
ing\TermReason\7724\Terminator: "HAM"
ing\TermReason\7724\Reason: 0x00000004
ing\TermReason\7724\CreationTime:  22 24 83 63 63 B0 DC 01
latileNotifications\41C64E6DA3675855:  01 00 04 80 44 00 00 00 50 00 00 00 00 00 00 14 00 00 00 02 00 30 00 02 00 00 00 00
:ings\S-1-5-21-164956077-1167442871-3309154556-1001\Microsoft.Windows.Apprep.ChxApp_cw5n1h2txyewy:  88 5B 1C 50 63 B0 DC 01 0
:ings\S-1-5-21-164956077-1167442871-3309154556-1001\\Device\HarddiskVolume3\Users\potato\Downloads\ShiginimaLauncher.exe:  E5
Settings\S-1-5-21-164956077-1167442871-3309154556-1001\Microsoft.Windows.Apprep.ChxApp_cw5n1h2txyewy:  88 5B 1C 50 63 B0 DC
Settings\S-1-5-21-164956077-1167442871-3309154556-1001\\Device\HarddiskVolume3\Users\potato\Downloads\ShiginimaLauncher.exe:
TWARE\Microsoft\Windows\CurrentVersion\Explorer\FeatureUsage\AppSwitched\C:\Users\potato\Downloads\ShiginimaLauncher.exe: 0x
TWARE\Microsoft\Windows\CurrentVersion\Explorer\FeatureUsage\TrayButtonClicked\MultitaskingButton: 0x00000001
```

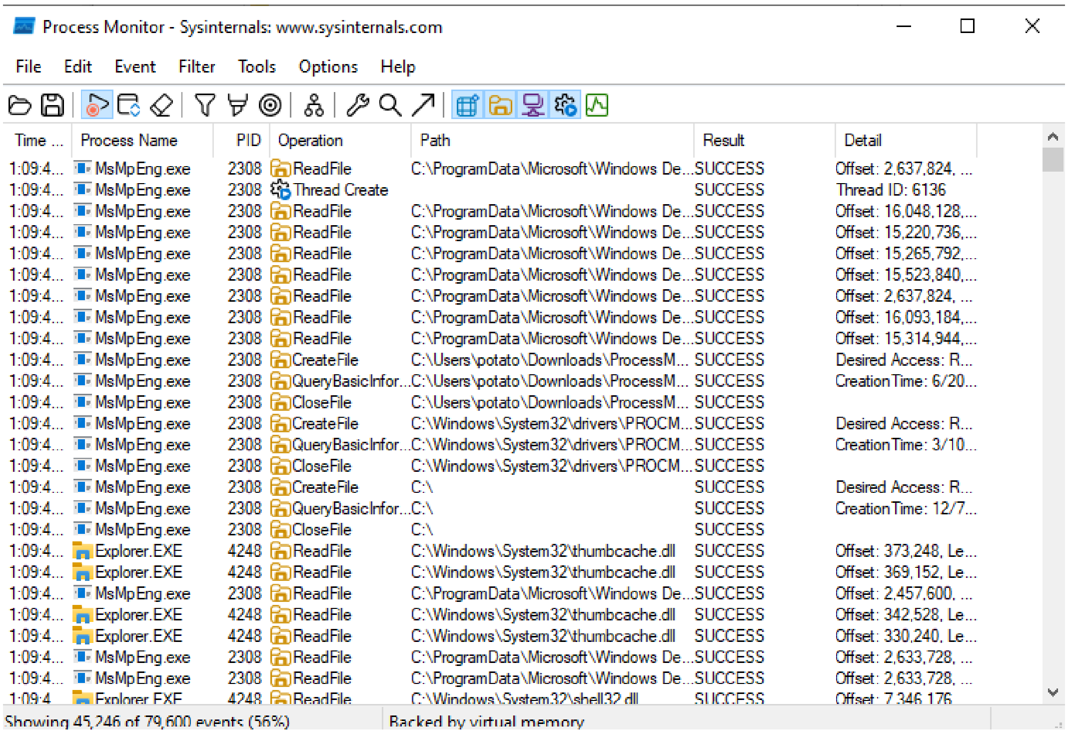
Uninstall (Delete .exe file)

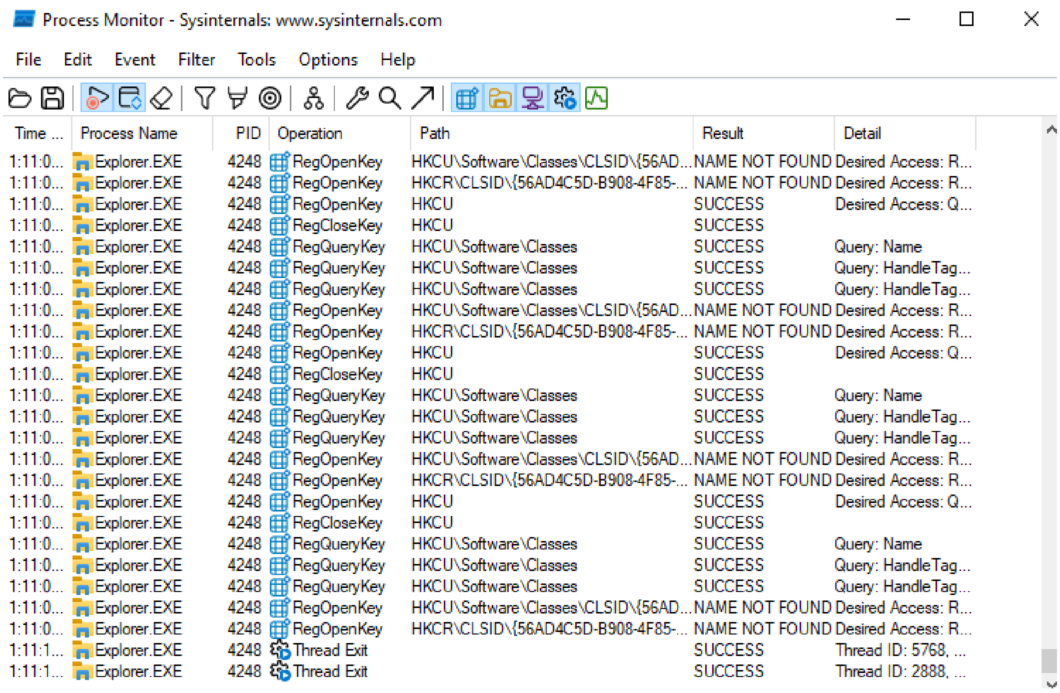
```
Regshot 1.9.0 x64 Unicode
Comments:
Datetime: 2026/3/10 08:01:58  ,  2026/3/10 08:03:43
Computer: DESKTOP-TNIER3R , DESKTOP-TNIER3R
Username: potato , potato

----------------------------------
Values modified: 9
----------------------------------
HKLM\SOFTWARE\Microsoft\Multimedia\Audio\Journal\Render:  53 00 57 00 44 00 5C 00 4
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
HKLM\SOFTWARE\Microsoft\Multimedia\Audio\Journal\Render:  53 00 57 00 44 00 5C 00 4
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\VFUProvider\StartTime:  7F 1D 42 2C
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\VFUProvider\StartTime:  B2 DE 05 50
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\418A073AA3BC34
```

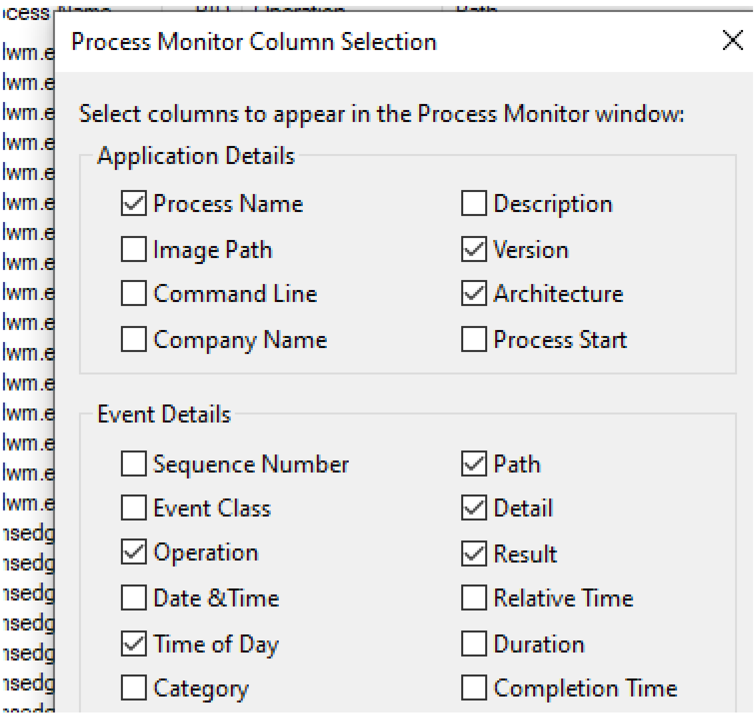
```
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0 00 00 00
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\
HKU\S-1-5-21-164956077-1167442871-3309154556-1001\SOFTWARE\Microsoft\Windows\

----------------------------------
Files deleted: 1
----------------------------------
C:\Users\potato\Downloads\ShiginimaLauncher.exe
```

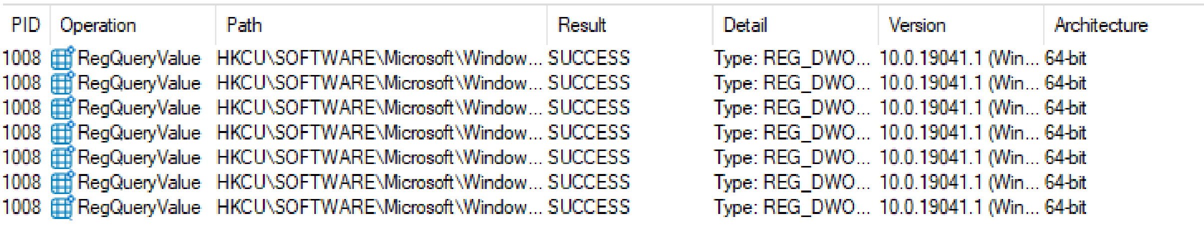
Lab Exercise 15.06: Process Monitor

1a-1e



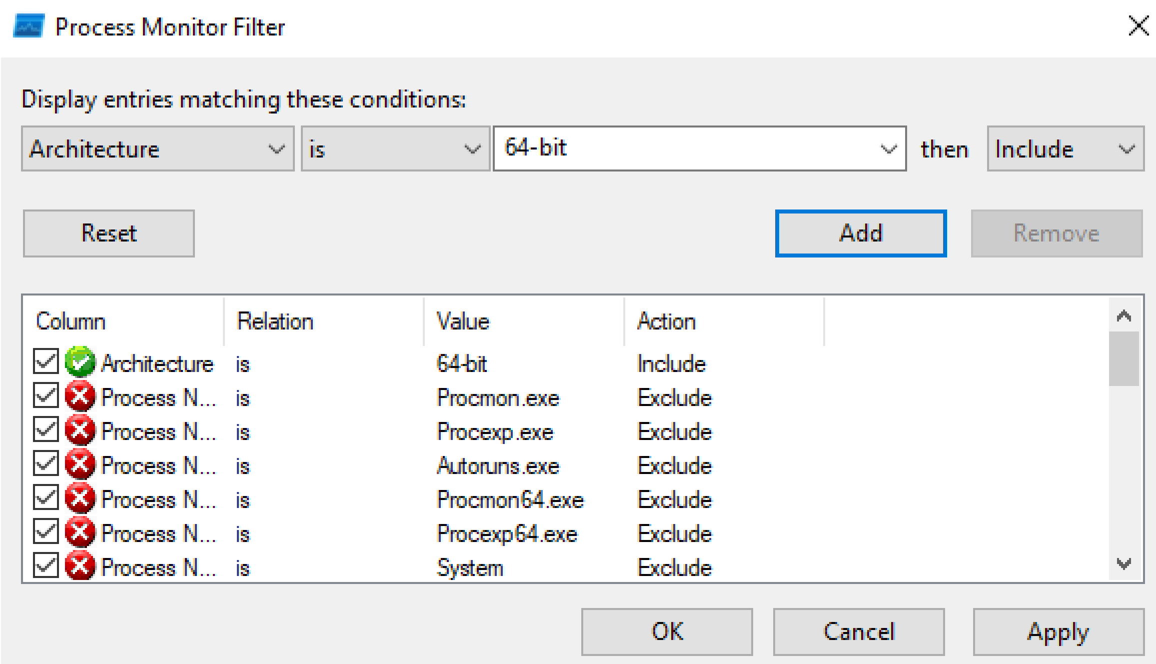




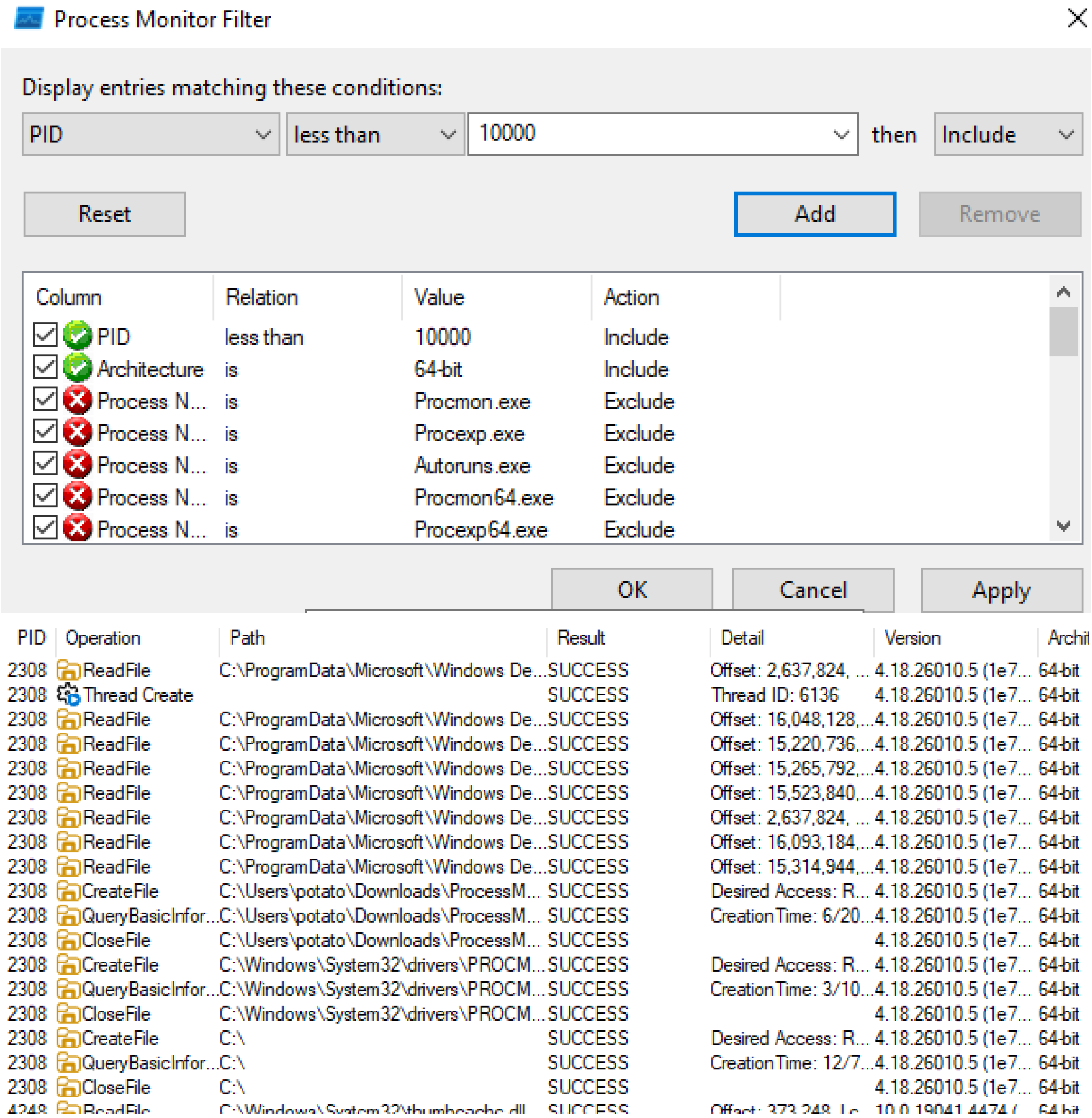


PID	Operation	Path	Result	Detail	Version	Architecture
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit
1008	RegQueryValue	HKCU\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_DWO...	10.0.19041.1 (Win...	64-bit

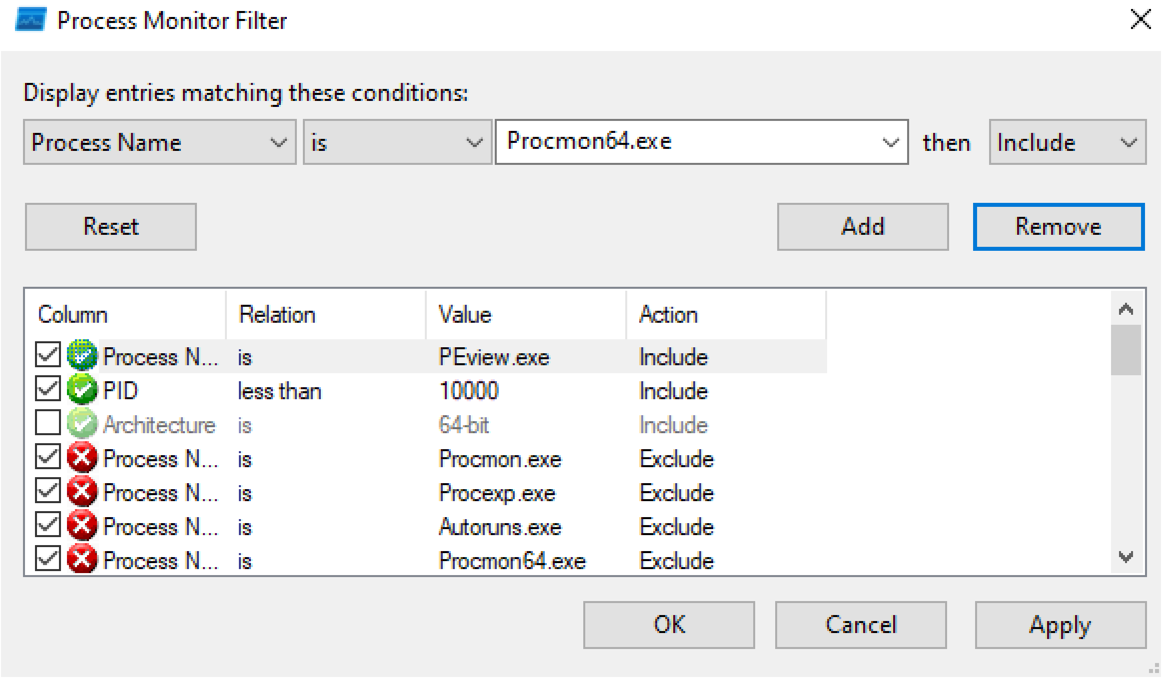
2a-2e

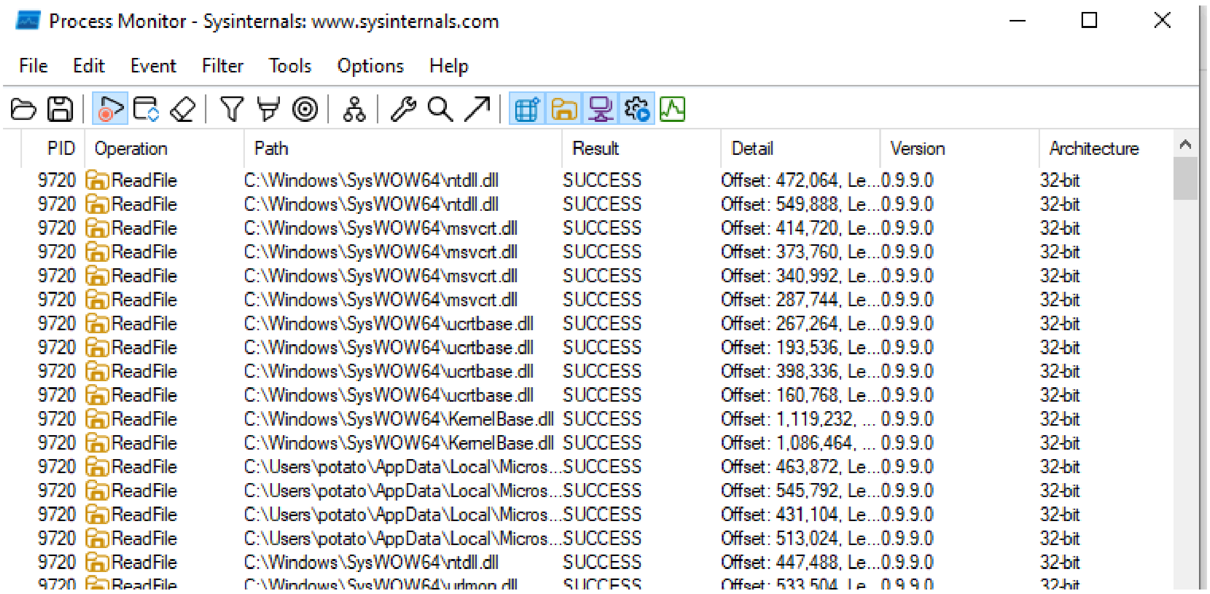


Add filter for selecting only operations that has PID number < 10000 and has Architecture 64-bit



Add filter for selecting only operations that has PID number < 10000 and has process name “PEViewer.exe”





We can use this to find out what processes and operations are being executed by an executable file, with given desired parameters.

Lab Exercise 15.07: ApateDNS

1a-1c

Command Prompt

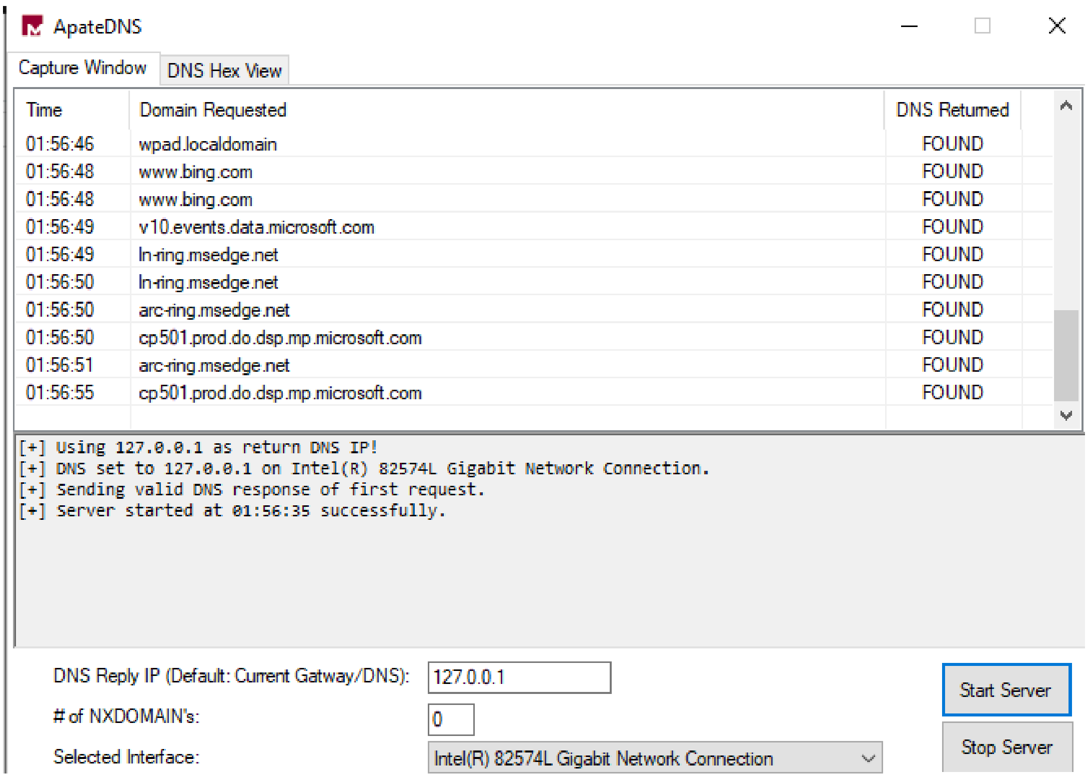
```
Microsoft Windows [Version 10.0.19044.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Users\potato>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-TNIER3R
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : localdomain

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-2D-86-3F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.102.128(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, March 10, 2026 1:46:19 AM
   Lease Expires . . . . . . . . . . : Tuesday, March 10, 2026 2:16:19 AM
   Default Gateway . . . . . . . . . : 192.168.102.2
   DHCP Server . . . . . . . . . . . : 192.168.102.254
   DNS Servers . . . . . . . . . . . : 192.168.102.2
   Primary WINS Server . . . . . . . : 192.168.102.2
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

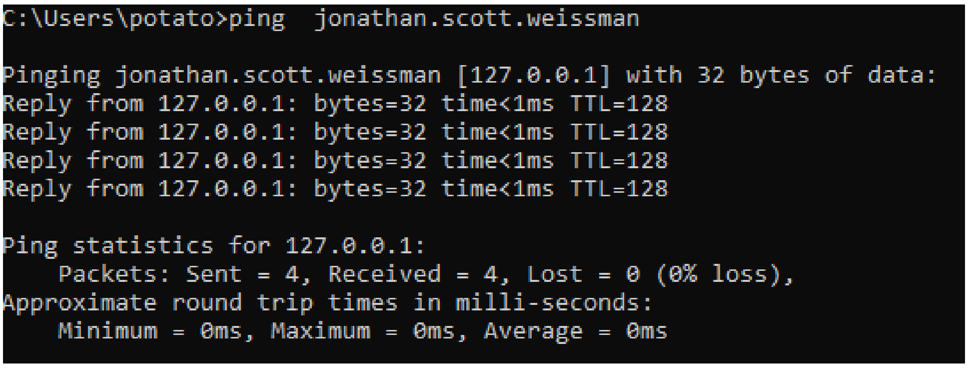
```
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5357           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49670          0.0.0.0:0              LISTENING
  TCP    192.168.102.128:139    0.0.0.0:0              LISTENING
  TCP    192.168.102.128:49680  4.213.25.241:443       ESTABLISHED
  TCP    192.168.102.128:49845  150.171.27.11:443      ESTABLISHED
```

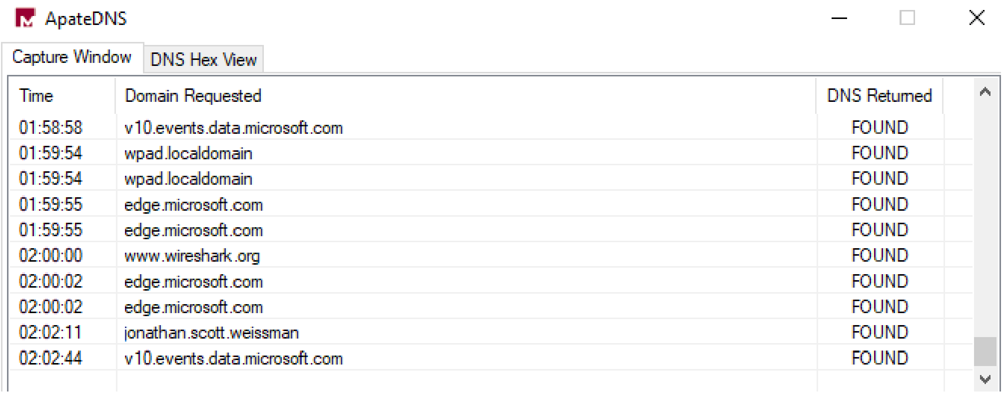
2a-2b

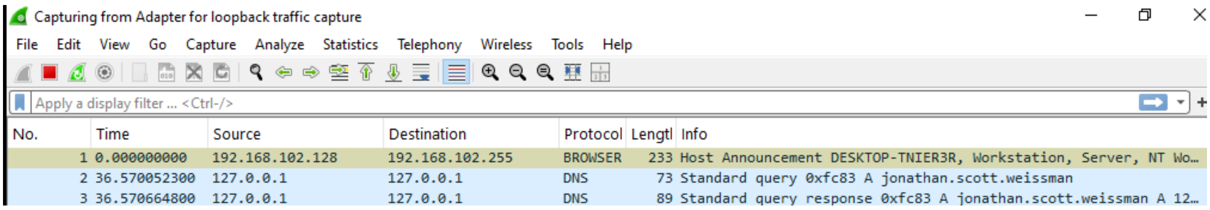
```
C:\Users\potato>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-TNIER3R
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : localdomain

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-2D-86-3F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.102.128(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, March 10, 2026 1:46:19 AM
   Lease Expires . . . . . . . . . . : Tuesday, March 10, 2026 2:26:35 AM
   Default Gateway . . . . . . . . . : 192.168.102.2
   DHCP Server . . . . . . . . . . . : 192.168.102.254
   DNS Servers . . . . . . . . . . . : 127.0.0.1
   Primary WINS Server . . . . . . . : 192.168.102.2
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

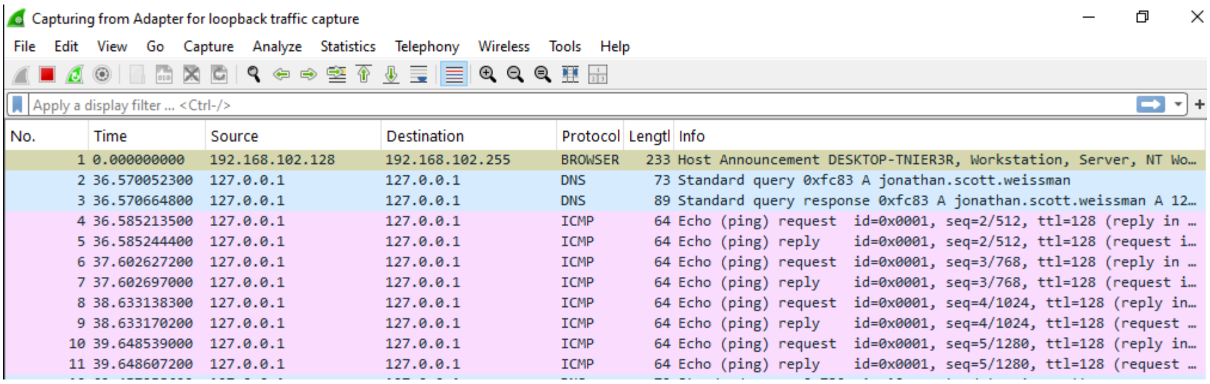
3a-3c



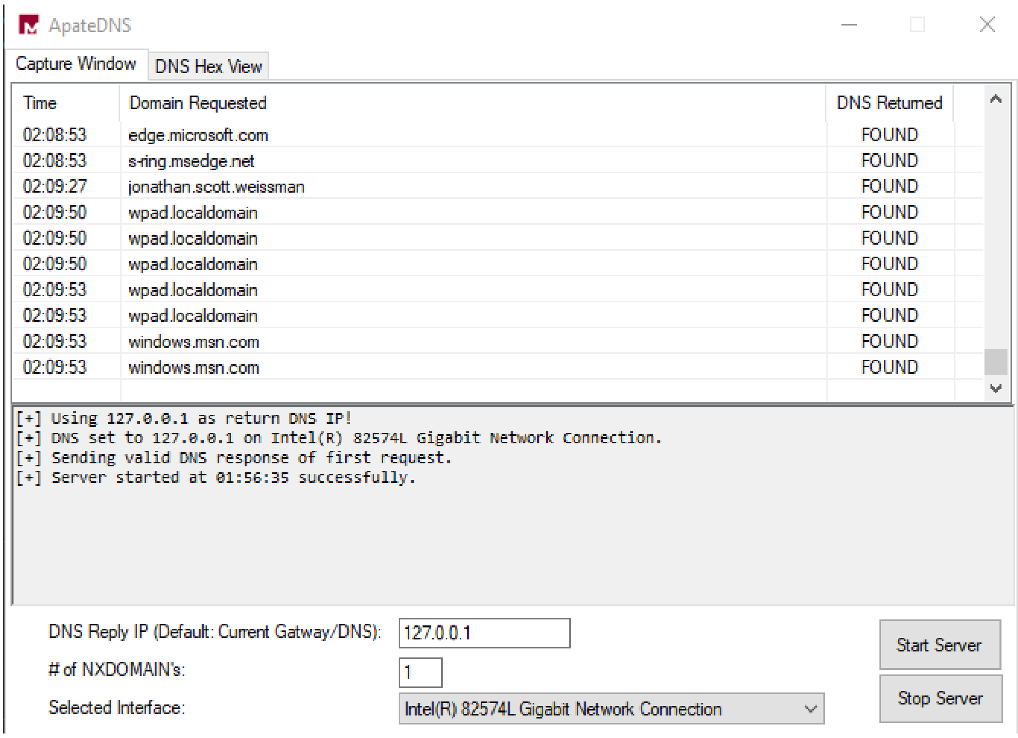

C:\Users\potato>ping jonathan.scott.weissman
Pinging jonathan.scott.weissman [127.0.0.1] with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms


ApateDNS
Capture Window
DNS Hex View
Time Domain Requested DNS Returned
01:58:58 v10.events.data.microsoft.com FOUND
01:59:54 wpad.localdomain FOUND
01:59:54 wpad.localdomain FOUND
01:59:55 edge.microsoft.com FOUND
01:59:55 edge.microsoft.com FOUND
02:00:00 www.wireshark.org FOUND
02:00:02 edge.microsoft.com FOUND
02:00:02 edge.microsoft.com FOUND
02:02:11 jonathan.scott.weissman FOUND
02:02:44 v10.events.data.microsoft.com FOUND

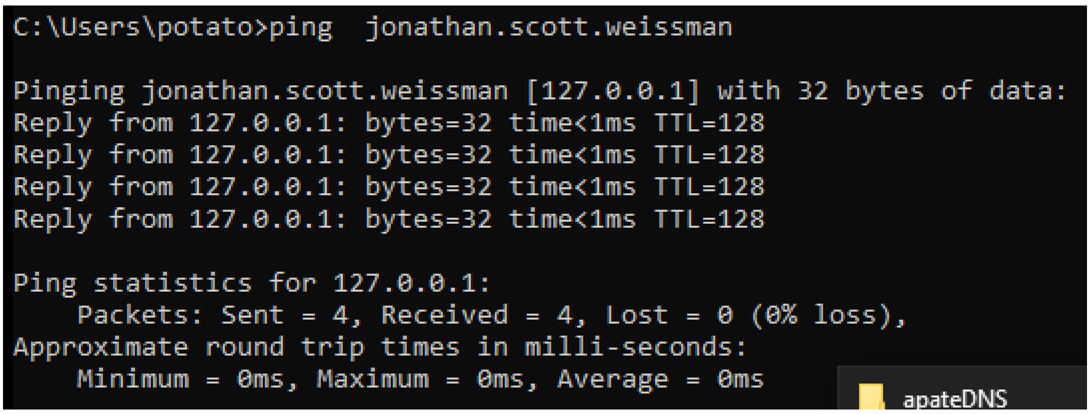

Capturing from Adapter for loopback traffic capture
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 192.168.102.128 192.168.102.255 BROWSER 233 Host Announcement DESKTOP-TNIER3R, Workstation, Server, NT Wo…
2 36.570052300 127.0.0.1 127.0.0.1 DNS 73 Standard query 0xfc83 A jonathan.scott.weissman
3 36.570664800 127.0.0.1 127.0.0.1 DNS 89 Standard query response 0xfc83 A jonathan.scott.weissman A 12…

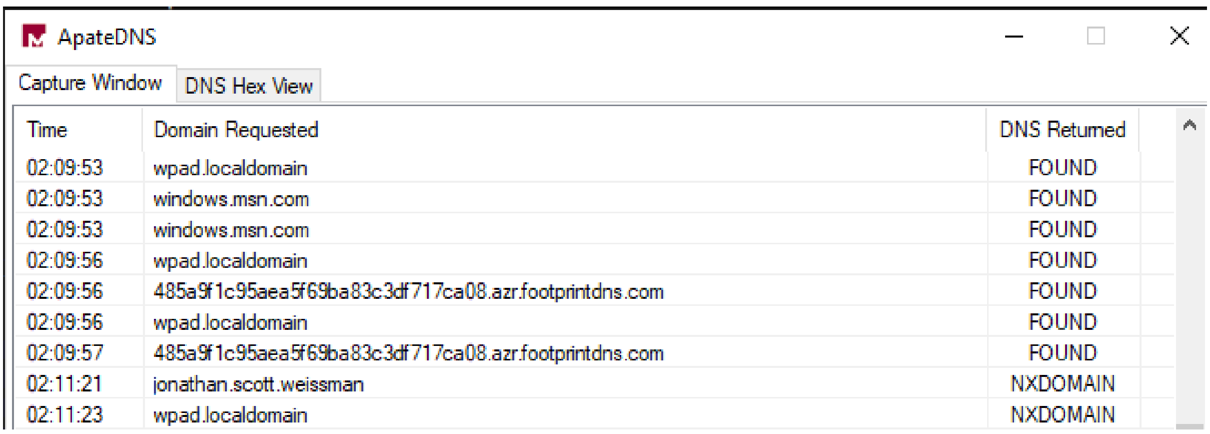

Capturing from Adapter for loopback traffic capture
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 192.168.102.128 192.168.102.255 BROWSER 233 Host Announcement DESKTOP-TNIER3R, Workstation, Server, NT Wo…
2 36.570052300 127.0.0.1 127.0.0.1 DNS 73 Standard query 0xfc83 A jonathan.scott.weissman
3 36.570664800 127.0.0.1 127.0.0.1 DNS 89 Standard query response 0xfc83 A jonathan.scott.weissman A 12…
4 36.585213500 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in …
5 36.585244400 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) reply id=0x0001, seq=2/512, ttl=128 (request i…
6 37.602627200 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in …
7 37.602697000 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) reply id=0x0001, seq=3/768, ttl=128 (request i…
8 38.633138300 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in…
9 38.633170200 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) reply id=0x0001, seq=4/1024, ttl=128 (request …
10 39.648539000 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) request id=0x0001, seq=5/1280, ttl=128 (reply in…
11 39.648607200 127.0.0.1 127.0.0.1 ICMP 64 Echo (ping) reply id=0x0001, seq=5/1280, ttl=128 (request …

4a-4c



NXDOMAIN (domain resolve):





Ping again, domain already resolved: FOUND

```
C:\Users\potato>ping  jonathan.scott.weissman

Pinging jonathan.scott.weissman [127.0.0.1] with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ApateDNS

Capture Window | DNS Hex View

Time	Domain Requested	DNS Returned
02:09:53	windows.msn.com	FOUND
02:09:53	windows.msn.com	FOUND
02:09:56	wpad.localdomain	FOUND
02:09:56	485a9f1c95aea5f69ba83c3df717ca08.azr.footprintdns.com	FOUND
02:09:56	wpad.localdomain	FOUND
02:09:57	485a9f1c95aea5f69ba83c3df717ca08.azr.footprintdns.com	FOUND
02:11:21	jonathan.scott.weissman	NXDOMAIN
02:11:23	wpad.localdomain	NXDOMAIN
02:11:30	jonathan.scott.weissman	FOUND
02:12:35	jonathan.scott.weissman	FOUND